



FIAS Frankfurt Institute
for Advanced Studies



BACHELOR THESIS

Self-Supervised Visual Learning for the Emergence of Sparse and Semantic Representations

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30 August 2026

Abstract

Self-supervised visual representations are dense by construction. The rotational invariance of the contrastive objective means that individual units carry no guaranteed semantic role: features spread across dimensions in arbitrary mixtures, and no unit is privileged over any other. This thesis investigates whether sparsity imposed during contrastive training is sufficient to produce unit-level semantic structure, and what controls how much emerges.

Three experiments address the question. The first characterizes the non-negativity gate of Non-Negative Contrastive Learning (**NNCL**) across the contrastive temperature. As temperature rises and gate-induced sparsity increases, individual units become more selectively responsive to object categories, as measured by a neuron-to-concept matching algorithm; SEPIN disentanglement, evaluated at the sparsity optimum, confirms the same structure. Both improvements come at no cost to downstream classification accuracy. This behaviour has a limit: a theoretical analysis of the margin bound identifies a critical temperature above which the gate induces dimensional collapse, and provides an explanation for where that threshold lies. Three factors are found to trigger collapse: high temperature, strong sparsity regularization, and easy positive pairs.

The second experiment isolates the mechanism. An explicit sparsity regularization applied to standard contrastive learning without any non-negativity constraint recovers category-level unit selectivity on par with the **NNCL** baseline and produces comparable disentanglement, without deactivating a single unit. The interpretability gains therefore arise from axis alignment rather than strict non-negativity. The two mechanisms are not equivalent: the gate produces hard binary activations and stronger identity matching, while the penalty offers stable training and a directly tunable degree of sparsity.

The third experiment tests whether the link holds outside the controlled setting. The same comparison is repeated on MIMo-Mirror, a temporal dataset in which positive pairs are drawn from temporally adjacent frames of an embodied agent’s visual experience. **NNCL** produces units that fire selectively over narrow joint-angle ranges. Standard contrastive learning encodes pose through continuous activation magnitude instead. Both representations carry unit-level pose information, but in different forms. The interpretability advantage of sparse codes is clearest when the target concept admits a binary encoding.

Across both datasets and both sparsity mechanisms, sparsity imposed during training is sufficient to produce unit-level semantic structure. The mechanism is the breaking of rotational invariance: any constraint that concentrates activation mass onto fewer dimensions removes the symmetry that prevents individual units from carrying stable semantic roles.

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1 Introduction

Before a vision model can classify, detect, or generate, it must transform its input into an internal representation: a learned encoding from which outputs are derived. What that representation encodes, and how it is organized, shapes both what the model can do and how well it generalizes beyond its training data.

Historically, the best representations were learned through supervised learning, where models are trained on large datasets of manually labelled examples. This approach is effective, but expensive (in time, cost, and scope). Labels must be created by hand for every task of interest, and because a model is optimized for the labels it was given, the resulting representation may capture little beyond what that particular task required.

Self-supervised methods sidestep the need for labels, learning directly from the structure of unlabelled data. But the resulting representations are usually dense, with features mixed together in ways that resist interpretation, and there is no clear way to attribute meaning to individual components. A model can perform well without any account of what its features encode.

This thesis investigates whether sparsity can change that. If a representation uses only a handful of features to describe each input, those features may carry distinct, recognizable meanings rather than blending together in ways that resist interpretation.

1.1 Background

The following sections establish what a contrastive representation is, how it is trained, and how the non-negativity constraint changes it, before defining what sparsity and semantic structure mean in this context.

1.1.1 Self-Supervised Learning

Self-supervised learning (SSL) is a way of training machine learning models without human-provided labels. Rather than learning to solve one specific task, it learns representations from the structure of the data itself. These representations capture the underlying factors of variation between samples and can be adapted to solve many downstream tasks.

Self-supervised learning is often compared to human learning, because both rely heavily on learning the structure of the environment without explicit labels. Humans do not need someone to label every object, event, or concept they encounter. Instead, patterns are observed and internal representations of the world are built without explicit instruction. Common self-supervised learning approaches are inspired by this ability to learn invariance. The same object can be recognized from different distances, orientations, lighting, and partial occlusion.

1.1.2 Contrastive Learning

Contrastive learning (CL) is a form of self-supervised learning in which the model learns which representations should be similar. The goal is to pull representations of the same object together, while pushing representations of different objects apart. This structures the representation space so that semantically similar samples cluster together while dissimilar ones are separated.

SimCLR [1] is the most popular contrastive learning framework. It works by taking an image and applying two random augmentations (cropping, colour distortion, flipping, blurring). The two augmented views of the same image form a positive pair, which should have the same representation.

The training objective is the NT-Xent loss [1], an instance of InfoNCE [2] adapted to the augmentation setting:

$$\ell(i, j) = -\log \frac{\exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_j)/\tau)}{\sum_{l=1}^{2N} \mathbb{1}_{[l \neq i]} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_l)/\tau)} \quad (1)$$

where $\mathbf{z}_i$ is the learned representation of example i , $\text{sim}(\mathbf{u}, \mathbf{v}) = \mathbf{u}^\top \mathbf{v} / (\|\mathbf{u}\| \|\mathbf{v}\|)$ is the cosine similarity, τ is a temperature parameter, and the sum runs over all $2N$ examples in the batch except i itself. Intuitively, the numerator rewards high similarity between the two views of the same image, while the denominator penalizes high similarity with all other images in the batch, pushing negative pairs apart.

Wang and Isola [3] formalize the two forces in the loss as geometric properties: alignment, which requires positive pairs to map to nearby points on the unit hypersphere, and uniformity, which requires representations to spread evenly across it. The numerator of NT-Xent optimizes alignment; the denominator optimizes uniformity. The two forces compete, and the learned representation settles at a trade-off between them.

A common failure mode of contrastive learning is representational collapse: the model learns to map all inputs to a lower-dimensional subspace [4], or even the same point in its most extreme form, satisfying alignment while compromising on uniformity. This results in low-rank solutions, where the encoder has learned a compressed, under-expressive representation. Input variation that is not captured within this lower-dimensional subspace is lost entirely and cannot be recovered downstream.

A vision encoder extracts a representation $\mathbf{h}$; a projection head maps it to $\mathbf{z} = g(\mathbf{h})$, the space where NT-Xent is applied. Counterintuitively, $\mathbf{h}$ transfers better to downstream tasks than $\mathbf{z}$ [1]. One explanation is that the contrastive loss forces $\mathbf{z}$ to be invariant to the augmentations used to form positive pairs, suppressing information the loss treats as noise but that downstream tasks may still find useful. The projector acts as a buffer that absorbs this invariance pressure, leaving $\mathbf{h}$ to retain broader information.

The temperature τ controls how sharply the model distinguishes between similar and dissimilar pairs. At low τ , the loss concentrates on the hardest negatives, driving strong separation and pushing representations towards a uniform spread across the hypersphere. At high τ , the gradient spreads more evenly, relaxing that pressure. Wang and Liu [5] formalize this as a uniformity-tolerance dilemma: a lower temperature drives stronger uniformity but makes the loss intolerant to semantically similar samples, treating them as hard negatives regardless of shared content.

1.1.3 Non-Negative Contrastive Learning

Non-Negative Contrastive Learning (NNCL) [6] is a modification of standard contrastive learning that enforces non-negativity on the projected representation $\mathbf{z}$ by applying a ReLU

activation after the projection head: $z = \text{ReLU}(g(\mathbf{h}))$. The contrastive loss and training procedure are otherwise unchanged.

The central theoretical motivation is that standard contrastive learning suffers from rotational invariance: if $\mathbf{z}$ is an optimal solution to the contrastive objective, then $\mathbf{R}\mathbf{z}$ for any orthogonal matrix $\mathbf{R}$ is equally optimal. The coordinate axes of the representation do not have any privileged meaning, so individual dimensions cannot be interpreted in isolation and disentanglement is unachievable in principle. Non-negativity breaks this symmetry. By constraining the representation space to the positive orthant, the separating force of the contrastive loss aligns the learned representation more closely along the coordinate axes, leading to sparse representations with selective dimensions. This mirrors a well-known property of Non-Negative Matrix Factorization (NMF) [7], a classical decomposition method that expresses data as a sum of non-negative components: non-negativity tends to produce representations that are sparse, selective, and aligned with semantically meaningful structure, where individual components correspond to recognizable parts rather than holistic mixtures.

Empirically, **NNCL** achieves roughly 90% sparsity per sample compared to near-zero for standard **CL**, with substantially improved disentanglement as measured by SEPIN, a per-dimension mutual information metric. Wang et al. [6] also report modest gains in downstream classification accuracy, though the mechanism behind the improvement is left unexplained.

A practical concern with ReLU in this setting is that dimensions pushed to zero during training may stop receiving gradient and never recover. **NNCL** addresses this with a straight-through estimator that uses the ReLU output in the forward pass but substitutes the GELU gradient in the backward pass, keeping gradient flow alive for inactive dimensions without breaking non-negativity.

1.1.4 Contrastive Learning Through Time

Contrastive Learning Through Time (CLTT) [8] is a self-supervised learning framework that uses time instead of augmentation to form positive pairs. The idea is inspired by how infants learn to recognize objects: by simply watching the world. Because objects persist over time, two frames that are close together in a video sequence almost certainly show the same object, just from a slightly different angle or position. CLTT uses this as a free learning signal.

It otherwise follows SimCLR closely, with one key difference: positive pairs are two successive frames from the same sequence, rather than two augmented views of a single image. Schneider et al. show that CLTT achieves near-supervised classification accuracy when frames close in time reliably depict the same object.

1.1.5 Sparse Representations

Sparse coding theory draws a useful distinction between two kinds of sparsity [9], [10]. *Population sparsity* describes how many units are active at once for a given input. A sparse code represents each example using only a few of its available units. *Lifetime sparsity* describes how often a single unit responds across all inputs. A selective unit fires only for a small fraction of them. The two reinforce each other: if each input activates only a few units, each unit can only see a small slice of all inputs. Population sparsity is the more directly useful property, as it ensures each input has a compact, distinct representation. The non-negativity gate realizes

this directly: negative activations are zeroed, leaving each unit either silent or active, so the binary split follows from the architecture itself.

Without a gate, sparsity can take a different form. Rather than a hard active/inactive split, activation mass is distributed unevenly: a small number of units carry most of the response while the rest remain near-zero. A penalty on distributed activation mass concentrates responses onto a small number of units while driving the rest towards zero, without requiring exact zeros [11]. In either case, a rarely active unit tends to specialize for a specific kind of input, and that selectivity makes it more interpretable [12].

This idea has a natural parallel in neuroscience. Neural coding in early visual cortex is sparse: at any moment, only a small fraction of neurons are active, and different inputs tend to activate different subsets [9]. Sparse activity is believed to emerge because it is efficient: fewer active neurons mean less overlap between codes, making patterns easier to tell apart and store without interference. A unit that fires constantly tells you little about which input is present; one that fires rarely signals something specific when it does.

1.1.6 Semantic Structure

Throughout this thesis, *unit* refers to a single dimension of an encoded representation vector.

A representation has *semantic structure* if its individual components correspond to recognizable concepts. In practice, this is measured at the unit level: a unit is considered semantically interpretable if it consistently activates for a specific visual concept (an object, a texture, a shape) and stays largely silent for everything else [12]. The cleaner that correspondence, the more a model’s internals can be described in terms a human can actually work with.

Nothing in standard training pushes a model to organize its features in a semantically clean way. A contrastive loss cares only that similar inputs end up nearby and different inputs end up far apart. Without explicit pressure towards structure, there is no reason for individual units to align with human-recognizable concepts. On top of that, geometry actively works against it. The superposition hypothesis [13] formalizes this: a network can represent far more than d independent features in a d -dimensional space by encoding each in a nearly-orthogonal direction. This lets a model pack more information into a fixed capacity, but the cost is interference: the more features are overlapped, the harder it becomes to read out any single one cleanly, and dense representations become superpositions of many more underlying sparse features than their dimensionality suggests.

1.2 Related Work

A consistent empirical finding, going back to computational neuroscience, is that sparse coding and interpretable structure co-occur in visual representations. Olshausen and Field [14] showed that learning a sparse code for natural images (with no supervision or labels) recovers Gabor-like filters closely matching the receptive fields of simple cells in primary visual cortex. Sparsity alone, applied to raw pixel statistics, recovers the feature vocabulary that early vision uses. Wright et al. [15] extend this to classical computer vision, demonstrating that representing images as sparse combinations of learned dictionary elements yields features that correspond to recognizable visual patterns and parts. Both results use a separately optimized dictionary rather than an end-to-end trained encoder, but together they establish the core

empirical claim this thesis builds on: sparsity and recognizable visual structure reliably go together.

The same logic has more recently been applied to recover interpretable structure from the dense activations of trained deep networks. Motivated by the superposition hypothesis (Section 1.1.6), Cunningham et al. [11] train sparse autoencoders (SAEs) on top of language model activations, finding that the recovered features are highly interpretable and correspond to specific, identifiable concepts. Bricken et al. [16] scale this up and show that the recovered features are genuinely monosemantic: a single SAE feature fires reliably for one recognizable concept across contexts. The consistent result is that decomposing a trained representation into a sparse basis can recover meaningful unit-level structure from representations that appear dense and entangled. This body of work bears directly on this thesis, but the relationship is one of contrast as much as precedent. SAEs impose sparsity after training, on top of a representation learned without any sparsity pressure, and can only decompose what the encoder already encodes. The question here is whether sparsity imposed during contrastive training shapes what the encoder learns to encode in the first place, producing unit-level semantic structure directly, without a separate decomposition step.

Adding a structural constraint to the contrastive objective is only useful if it does not destabilize training. Barlow Twins [17] and VICReg [18] show that it does not: both methods address dimensional collapse by penalizing correlation between feature dimensions, with Barlow Twins pushing the cross-correlation matrix of two augmented views towards the identity and VICReg adding variance and covariance penalties alongside an invariance term. Neither method targets sparsity or interpretability, but both demonstrate that objectives that impose geometric structure on z beyond the contrastive loss are training-stable and downstream-beneficial. Hua et al. [19] make the benefit more precise: feature decorrelation alone improves linear probe accuracy, linking structural constraints directly to representation quality. Together, these results establish the precedent that shaping the geometry of the projected representation is viable and productive. Xu et al. [20] demonstrate that a Hoyer sparsity penalty can be combined with an InfoNCE-style contrastive loss directly, improving contradiction retrieval in NLP; the combination is stable and the sparsity term acts as a reliable control lever, consistent with the geometric argument above. Sparsity is a different kind of geometric constraint, but it belongs to the same family of moves.

The closest prior work to this thesis is Non-Negative Contrastive Learning (NNCL) [6], described in Section 1.1.3. NNCL establishes that a non-negativity gate is compatible with contrastive training and that the resulting representations are measurably more structured. What it leaves open is what individual units actually respond to. Interpretability is measured through disentanglement metrics and feature selection benchmarks, not through any analysis of unit-level selectivity. Whether the sparsity the gate produces is sufficient to generate semantic structure at the level of individual units, what controls how much emerges, and how the gate interacts with the temperature that governs that sparsity are the questions this thesis takes up.

1.3 Aims and Objectives

Contrastive representations transfer well across downstream tasks but offer little insight into what they actually encode. Their geometry is rotationally invariant (Section 1.1.3), so individual dimensions carry no privileged meaning and features spread across many dimensions simultaneously [6], [13].

Sparsity breaks this invariance (Section 1.1.5), giving individual units a privileged role. Whether this produces genuine semantic structure (Section 1.1.6) in a contrastive setting, and what controls how much emerges, is the question this thesis pursues.

The question is posed at the level of individual units, where the effect of sparsity becomes directly measurable:

Does unit-level semantic structure emerge with sparsity in contrastive visual representations, and what controls that emergence?

The question is pursued through three objectives, each building on the previous one.

- 1. Does structure emerge with the sparsity of the non-negativity gate, and over what range?** Section 3 sweeps the contrastive temperature, which varies the sparsity the gate produces without changing the objective. The sparsity studied here is only the sparsity the gate happens to produce, and it cannot be set directly. The upper limit found there is not incidental: it bounds the range over which the gate is usable at all, and motivates the next objective.
- 2. How much of the effect is non-negativity, and how much is axis alignment alone?** The gate does two things at once: it confines representations to the positive orthant, and it aligns them with the coordinate axes. An L_1 penalty produces the weaker of the two, axis alignment without the orthant constraint, and does so through a coefficient that can be set directly. Section 4 applies the penalty with and without the gate to separate the contributions of the two mechanisms.
- 3. Does the structure hold in a naturalistic learning setting?** The first two objectives train models to learn rotation-invariant object identities, with unit-level selectivity scored against a fixed set of known classes. Section 5 repeats the comparison on MIMo-Mirror, where positive pairs are formed from temporal continuity, as in infant learning, and the generating factors are continuous rather than discrete. Persistence of the relation here reinforces the parallel with sparse coding in biological vision.

The scope is intentionally narrow. This thesis studies only visual representations learned through contrastive objectives; findings may not transfer to language models, generative encoders, or other self-supervised frameworks. Two sparsity-inducing mechanisms are compared: the non-negativity gate as implemented in NNCL and an explicit sparsity regularization applied to standard contrastive learning. Other sparsity mechanisms, such as sparse autoencoders applied post-hoc, are used as conceptual reference but not evaluated directly. Interpretability is operationalized at the unit level, measuring whether individual representation dimensions respond selectively to semantic categories or generating factors. Downstream

task performance is reported to compare the general information content of representations; improving downstream performance is not a primary goal.

2 Methods

This chapter describes the datasets, architectures, and training procedures used to investigate sparse and semantically structured representations, along with the evaluation protocols. The implementation is available at <https://github.com/fmhaus/ssl-emergence-sparse-semantic>[◦].

2.1 Experimental Overview

[Section 3](#) trains models to learn rotation-invariant representations of object identities on a dataset of objects recorded from many rotation views ([Section 2.2.1](#)). A non-negativity gate ([Section 1.1.3](#)) is added to the contrastive objective, producing sparse representations. The central question is whether this sparsity gives rise to unit-level semantic structure: do individual units become selectively responsive to specific object identities or categories, and how strongly?

[Section 4](#) builds on this by testing an alternative route to sparsity. Rather than a hard gate that switches units on or off, a training loss penalty is used to incentivize concentration of activation mass onto a small number of units ([Section 1.1.5](#)), without forcing any unit to be exactly silent. The experiment asks whether this softer form of sparsity is sufficient to produce the same kind of unit-level selectivity, and what the gate contributes beyond sparsity alone.

[Section 5](#) applies the findings to a naturalistic dataset, in which an agent observes a scene from successive viewpoints over time rather than from controlled rotation views ([Section 2.2](#)). Positive pairs are formed from temporal continuity, mirroring how infants acquire visual representations from simply watching the world. The goal is to test whether the link between sparsity and unit-level interpretability extends beyond the controlled setting, drawing a parallel to sparse coding in biological vision.

2.2 Datasets

Two datasets are used across the experiments: Revolve Toys4k (RT4K), which provides controlled viewpoint variation for the identity-learning setting, and MIMo-Mirror, an embodied dataset with temporal continuity for the naturalistic setting.

2.2.1 Revolve Toys4k (RT4K)

This dataset contains 3574 objects extracted from the Toys4k dataset [\[21\]](#), previously used in a similar self-supervised setting [\[22\]](#). The objects are recorded in two-degree steps of yaw-axis rotation at 128×128 pixel resolution, yielding 180 unique views per object.

One of 104 categories is assigned to each object, with the same set of categories appearing in both splits. The distribution of objects per category is uneven, with each category having 34 objects on average, but as few as 8 or as many as 186.

Positive pairs are drawn from a fixed subset of 512 RT4K objects, chosen to balance category representation and held fixed across all experiments. The 180 rotation views of each object are split by index parity: the 90 even-indexed views (0, 2, ..., 178) form the training split used exclusively to construct positive pairs, and the 90 odd-indexed views (1, 3, ..., 179) form the evaluation split. No view appears in both. This interleaved split tests whether the learned representation generalizes to samples not seen during training, while ensuring that every

evaluation view lies angularly between two training views. A contiguous held-out slice would instead test generalization to an entire side of the object that training never covered, which is a qualitatively harder out-of-distribution problem and not the focus here.

The dataset therefore holds data out along two axes, and the evaluations in [Section 2.4](#) use them differently. Identity labels exist only for the 512 training objects, so every evaluation scored against identity uses those objects and their odd-indexed views. Category labels are defined for all 3574 objects, so the category linear probe ([Section 2.4.4](#)) uses the objects outside the 512-object subset, which contribute no positive pair during training.



Figure 1: Two RT4K objects (piano, donut) each shown from two rotation views, illustrating the yaw-axis variation available per object.

2.2.2 MIMo-Mirror

This dataset was generated for evaluating Contrastive Learning Through Time [\[8\]](#) on visual input from MIMo, a multimodal infant model [\[23\]](#). MIMo is placed in front of a mirror and performs random arm movements. All other joints are fixed, so the only source of visual variation across frames is arm configuration.

MIMo observes its own mirror reflection through a single (left) eye camera with a 45-degree field of view. The head is held at a fixed pose (swivel -5° , tilt 15°).

At each timestep, the dataset contains the following data:

- A 64×64 RGB image of MIMo’s view through its left eye camera.
- An 8-dimensional vector describing the joint angles (shoulder horizontal, shoulder abduction, shoulder rotation, elbow flexion; left and right each).
- A 14-dimensional motor action vector that drives the movement between frames.

The motor actions are sampled randomly and applied to the arms after each frame. The simulation is then advanced 50 physics ticks (equal to $0.5s$). This yields an effective frame rate of 2 FPS (frames per second).



Figure 2: Sample frames from the MIMo-Mirror dataset. From left to right: default pose; random arm pose; view partially occluded by elbow; view almost fully covered by arm.

2.3 Model and Training

The model consists of a ResNet-18 backbone [24] that produces the representation $\mathbf{h}$ used for all downstream evaluation and a projection head that produces the contrastive embedding $\mathbf{z}$, to which the loss is applied. The architecture and base training setup are shared across all experiments; the two sparsity mechanisms differ between them.

2.3.1 Backbone Encoder

ResNet-18 [24] is an 18-layer convolutional neural network with residual connections that allow gradients to flow through deep stacks without vanishing. It produces the backbone representation $\mathbf{h} \in \mathbb{R}^{512}$ via global average pooling. For MIMo-Mirror, the default 7×7 stem is replaced with a 3×3 convolution at stride 1 without max pooling, following the small-image variant from [24], to avoid over-downsampling its 64×64 inputs. RT4K uses the default stem unchanged.

2.3.2 Projection Head

The backbone output $\mathbf{h} \in \mathbb{R}^{512}$ is passed through a two-layer multilayer perceptron (MLP) projection head, following the SimCLR design [1]. The MLP consists of two fully connected layers with batch normalization and ReLU in between. The bias terms are omitted, as batch normalization renders them redundant. Two optional components can be appended at the output: a batch normalization layer for training stability (Section 3.1); and the non-negativity gate from NNCL (Section 2.3.4), used in Section 3. The contrastive loss is applied to the resulting $\mathbf{z}$.

Two projector configurations are investigated. The first, referred to as the *small projector*, uses a hidden dimension of 4096 and an output dimension of 512, matching the number of objects in the RT4K (Section 2.2.1) training set: the intent being that the projector may learn to allocate one dimension per identity. The second, referred to as the *large projector*, uses a hidden dimension of 16384 and an output dimension of 4096, providing substantially more capacity in both the intermediate and output representations, with the hypothesis that a larger projector may make it easier to form sparse codes at $\mathbf{z}$.

2.3.3 Hyperparameters

Training runs for 100 epochs with a batch size of 256, using the AdamW optimizer with a learning rate of 3×10^{-4} and weight decay 10^{-4} . The learning rate follows a warmup-cosine schedule. It increases linearly during the first 10 epochs of training to stabilize early training and then decreases in a cosine shape. Mixed-precision training uses bfloat16 throughout. The contrastive objective is the NT-Xent loss Section 1.1.2 applied to the projector output $\mathbf{z}$, with temperature τ treated as a hyperparameter. Every configuration is trained with three different random seeds to record training variance and stability.

2.3.4 Non-Negativity Gate

As described in Section 1.1.3, NNCL [6] constrains the projector output to be non-negative, which induces sparsity through the contrastive objective. This section describes how the gate is implemented and how NNCL solves the gradient problem that a naive ReLU gate creates.

The gate is applied element-wise to the final linear layer’s output $\tilde{\mathbf{z}} \in \mathbb{R}^D$:

$$z^+ = \text{ReLU}(\tilde{z}) \quad (2)$$

A plain ReLU gate produces exactly zero for any negative pre-activation, which also zeroes out its gradient. Once a unit’s pre-activation is pushed negative by the contrastive loss, it receives no gradient and cannot recover, permanently silencing it.

NNCL addresses this with a reparameterization that keeps ReLU in the forward pass for strict non-negativity, but routes gradients through GELU [25] instead. Because GELU is smooth and has non-zero gradient for negative inputs, units that have been pushed below zero still receive a signal and can recover. In PyTorch, this is expressed as:

```
z = relu(z) + gelu(z) - gelu(z.detach())
```

In the forward pass, `gelu(z) - gelu(z.detach())` evaluates to zero, so the output is exactly `relu(z)` and non-negativity is preserved. In the backward pass, `.detach()` stops gradient flow through the second GELU term, leaving only the gradient of the first GELU, which replaces the ReLU gradient without changing the forward value.

2.3.5 Sparsity Regularization

In Section 4, a regularization term that rewards sparser representations is added to the contrastive objective [20]:

$$\mathcal{L} = \mathcal{L}_{\text{NCE}} + \lambda(1 - S_H(z)) \quad (3)$$

where λ controls the strength of the regularization and S_H is the Hoyer sparsity measure (Section 2.4.1). Since the projector output is unit-normalized ($\|z\|_2 = 1$), the penalty simplifies to

$$1 - S_H(z) = \frac{\|z\|_1 - 1}{\sqrt{D} - 1} \quad (4)$$

Minimizing it is equivalent to minimizing $\|z\|_1$, which pushes activations towards the coordinate axes. The normalization by $\sqrt{D} - 1$ makes the penalty independent of projector width, so the same λ applies comparable sparsity pressure across both projector configurations. Unlike the non-negativity gate, which produces hard zeros by deactivating units entirely, the penalty encourages axis alignment without hard deactivation: units remain active but are pushed towards sparser directions.

2.4 Evaluation

The evaluation framework has four main parts. Sparsity analysis compares how sparse different models actually are. Interpretability metrics test the link between sparsity and semanticity: does sparsity actually induce unit-level semantic structure? Representational geometry checks that the feature space is structurally healthy and diagnoses failure modes like dimensional collapse. Finally, downstream evaluation measures how useful the learned representations are for classification.

2.4.1 Sparsity Analysis

Sparse activations are the central objective of this thesis. Following the terminology of [9], two complementary views of sparsity are reported.

Lifetime sparsity characterizes a single unit across many samples: it is the fraction of samples for which that unit is inactive, so a unit with high lifetime sparsity is one that responds only rarely.

Population sparsity characterizes a single input across many units: it is the fraction of units that are inactive for that sample, so a highly sparse input activates only a small subset of the representation. Both quantities are reported as means over the evaluation set, and their joint distribution gives a more complete picture than either alone.

Population and lifetime sparsity measure hard sparsity: they count exact zeros, so they are only meaningful when units can be deactivated entirely, as the non-negativity gate does. Hoyer sparsity [26] complements them by quantifying soft sparsity (Section 1.1.5): how concentrated the activation mass is, regardless of whether any unit is exactly zero. It applies equally to gate-induced codes and to L_1 -regularized ones, making it the only sparsity measure directly comparable across both variants. For a vector $\mathbf{h} \in \mathbb{R}^D$, it is defined as:

$$S_H(\mathbf{h}) = \frac{\sqrt{D} - \|\mathbf{h}\|_1 / \|\mathbf{h}\|_2}{\sqrt{D} - 1} \quad (5)$$

The ratio $\|\mathbf{h}\|_1 / \|\mathbf{h}\|_2$ achieves its minimum 1 when the vector has exactly one non-zero entry and its maximum $\sqrt{D}$ when all entries are equal. Hoyer sparsity rescales this so that a value of 1 corresponds to a fully localized vector and 0 to a completely dense one. It is reported per sample and averaged over the evaluation set, giving an axis-alignment proxy independent of activation scale.

Finally, the dead-neuron percentage is reported separately from the two sparsity averages. A unit is classified as dead when it has 100% lifetime sparsity, meaning it never activates for any evaluation sample. Dead neurons inflate the mean lifetime sparsity without contributing any information: a representation that achieves high average sparsity partly by silencing units permanently is not more interpretable than a denser one; it simply has fewer effective dimensions.

2.4.2 Interpretability

Interpretability metrics try to measure whether individual units become selectively responsive to specific identities or measurable concepts.

2.4.2.1 Neuron-Level Matching

Neuron-level matching tests whether individual units in the representation have become selectively responsive to specific identities or categories. The procedure operates on the held-out evaluation views described in Section 2.2.1, which are disjoint from the views used to form positive pairs. Let $\mathbf{h}^{(i)} \in \mathbb{R}^D$ denote the representation of sample i and let $c(i) \in \{1, \dots, C\}$ denote its class label, where C is the number of classes and D is the representation dimension.

For each unit d , the activations are first standardized across the dataset:

$$\tilde{h}_d^{(i)} = \frac{h_d^{(i)} - \mu_d}{\sigma_d} \quad (6)$$

where μ_d and σ_d are the mean and standard deviation of unit d over all evaluation samples. The affinity of unit d for class c is then the expected normalized activation over samples of that class:

$$A_{dc} = \mathbb{E}[\tilde{h}_d \mid c(i) = c] \quad (7)$$

This produces an affinity matrix $\mathbf{A} \in \mathbb{R}^{D \times C}$. Intuitively, it measures how strongly and reliably a unit is active for a given class. The normalization allows comparing activation strength independently of a unit’s activation scale and shift. The Hungarian algorithm finds the globally optimal one-to-one assignment $m : \{1, \dots, C\} \rightarrow \{1, \dots, D\}$ of classes to units:

$$m^* = \arg \max_m \sum_{c=1}^C A_{m(c),c} \quad (8)$$

This gives every class exactly one matched unit, and every unit is assigned to at most one class. Matching is used to test unit-level selectivity for two kinds of concept: the 512 training object identities and the 104 RT4K categories. Both run on the 512 training objects and their odd-indexed views, unlike the category linear probe (Section 2.4.4), which uses the objects outside that subset. The two probe different aspects of the learned code. Identity is directly encoded in the training signal: positive pairs are built from views of the same object, so the objective explicitly rewards representations that bring each identity together. Whether individual units become selective for identities is then a question about the geometry of that learned code. Category is never directly supervised; objects within the same category share visual properties but are treated as different identities in training, so any category structure that emerges reflects visual similarity learned implicitly rather than anything the objective was asked to produce.

Once m^* is fixed, each sample is classified by assigning it to the class whose matched unit has the highest normalized activation:

$$\hat{c}(\mathbf{h}) = \arg \max_c \tilde{h}_{m^*(c)} \quad (9)$$

The affinity matrix and the accuracy are computed on the same evaluation views.

This single-neuron accuracy serves as a disentanglement proxy: it tests whether each identity or category is encoded in a single dedicated unit. Under a perfectly disentangled local code, the matched unit alone should be sufficient to identify its class, and single-neuron accuracy should approach the full linear probe accuracy (Section 2.4.4). A large gap between the two indicates that the concept is distributed across multiple dimensions rather than localized, even if average sparsity is high.

2.4.2.2 Per-Unit Correlation

Neuron-level matching requires a discrete ground-truth factor, like identity and category labels in Section 2.2.1. On MIMo-Mirror (Section 2.2.2), the factors are the eight individual

joint angles, which are continuous, so selectivity has to be measured against a real-valued target. Let $h_d^{(i)}$ denote the activation of unit d on sample i and let $y_j^{(i)}$ denote the value of factor j on the same sample. The Pearson correlation between the two is:

$$r_{dj} = \frac{\text{Cov}(h_d, y_j)}{\sigma_{h_d} \sigma_{y_j}} \quad (10)$$

where the covariance and standard deviations are taken over the evaluation set. Its square r_{dj}^2 is the fraction of the variance of factor j that a single unit explains linearly. The measure is invariant to scale and shift of both arguments. For each factor, the best unit is reported:

$$R_j^2 = \max_d r_{dj}^2 \quad (11)$$

This is the continuous analogue of matching: it asks whether any individual unit carries the factor on its own, rather than whether the factor is decodable from the representation as a whole.

The measure is deliberately linear, which makes it an unsuitable test for sparse representations. A unit that is inactive on most samples cannot track a continuous factor linearly, even when it carries information about that factor.

2.4.2.3 Activation Specificity

Linear correlation is only one form in which a unit can encode a factor. A unit may instead act as a detector, firing only when the factor falls inside a narrow range and remaining silent otherwise. Such a unit carries precise information about the factor whenever it is active, while its correlation across the full dataset can be near zero.

To capture this, the spread of a factor is compared between all samples and only those samples on which the unit is active, meaning its activation is strictly positive. Denote by σ_j the standard deviation of factor j over the whole evaluation set and by $\sigma_{j|d}$ the same quantity restricted to the samples on which unit d is active. Activation specificity is defined as:

$$S_{dj} = 1 - \frac{\sigma_{j|d}}{\sigma_j} \quad (12)$$

A value near 0 or below means that knowing the unit is active says nothing about the factor: the spread is as wide as over the whole set, or wider. A value near 1 means that activity of the unit pins the factor down to a narrow range, so the unit functions as a detector for one region of the factor space. The mean of the factor over the same samples, $\mu_{j|d}$, identifies which region that is and thus what pose a highly selective unit responds to.

Activation specificity treats a unit as a binary signal: what matters is whether it fires, not how strongly. A unit is interpretable in this sense if its activity alone narrows the factor down to a small range.

Both metrics are biased. Correlation favours dense representations, since a linear fit is poor under sparsity. Activation specificity favours a sparse code with many inactive units, since a dense model has no incentive to attribute meaning to a positive activation. A direct compar-

ison of the two should therefore be treated with care, since different results are expected by construction.

2.4.2.4 Separated Information (SEPIN)

SEPIN (Separated Information), introduced by Do and Tran [27], is an unsupervised disentanglement metric. $\text{SEPIN}_d = I(x; f(x)) - I(x; f_{-d}(x))$ measures how much each single dimension individually contributes to the information captured by the whole representation. The mutual information is approximated by the InfoNCE lower bound [2], which for a batch of N samples is:

$$I(x; f(x)) \geq \log N - \mathcal{L}_{\text{NCE}(f)} \quad (13)$$

Analogously, $I(x; f_{-d}(x))$, the mutual information of the representation without dimension d , is bounded with dimension d masked out:

$$I(x; f_{-d}(x)) \geq \log N - \mathcal{L}_{\text{NCE}(f_{-d})} \quad (14)$$

Both terms are computed on the held-out evaluation views using the frozen encoder. The constant $\log N$ cancels in the difference, so SEPIN_d is estimated as the increase in NCE loss caused by masking dimension d :

$$\text{SEPIN}_d \approx \mathcal{L}_{\text{NCE}(f_{-d})} - \mathcal{L}_{\text{NCE}(f)} \quad (15)$$

Strictly, a difference of two lower bounds is not itself a bound on the difference; the estimate is used as a relative measure of per-dimension contribution rather than as an absolute quantity.

$\text{SEPIN}@k$ aggregates the top- k most informative dimensions:

$$\text{SEPIN}@k = \frac{1}{k} \sum_{i=1}^k \text{SEPIN}_{r_i} \quad (16)$$

where $r_1, r_2, \dots, r_k$ are the indices of dimensions sorted by SEPIN_d in descending order.

The shape of $\text{SEPIN}@k$ at different k values gives insight into how information is encoded in the learned representation.

- $\text{SEPIN}@1$ reports the single most informative dimension: a high value indicates that one dimension alone carries a high proportion of the total information.
- $\text{SEPIN}@10$ and $\text{SEPIN}@100$ capture increasingly broad slices of the top dimensions, revealing how quickly the per-dimension contribution drops off: a steep drop from $\text{SEPIN}@1$ to $\text{SEPIN}@10$ indicates that information is concentrated in very few dimensions, while a gradual decline suggests a more spread-out representation.
- $\text{SEPIN}@all$ is the mean contribution averaged across all D dimensions. A low value indicates most dimensions are individually expendable, because information is either encoded redundantly or the dimensions contribute no information. A higher value can be interpreted as most dimensions contributing meaningful information individually. At the projector widths used here it sits close to zero for every model, and can fall marginally below it because

each per-dimension term is a difference of two separately estimated losses, so it is reported for completeness rather than as a basis for comparison.

Because backbone representations $\mathbf{h}$ are not normalized to the unit sphere like the projector output $\mathbf{z}$, individual dimensions can differ substantially in scale. Feeding raw features into the InfoNCE estimator would cause high-variance dimensions to dominate, making the leave-one-out comparison unreliable. To correct for this, each dimension is z-score normalized across the evaluation set before computing $\mathcal{L}_{\text{NCE}}$:

$$\tilde{h}_d = \frac{h_d - \mu_d}{\sigma_d} \quad (17)$$

One practical concern is that the backbone $\mathbf{h}$ is not directly trained to optimize $\mathcal{L}_{\text{NCE}}$. This means the reliability of using it as an estimator for the mutual information is uncertain. However, SEPIN does not report the absolute mutual information, but the difference when masking out individual dimensions. This difference should still provide insight into the marginal contribution of each dimension. Empirical evaluation also indicates that backbone $\mathbf{h}$ and projector $\mathbf{z}$ have similar structure under downstream evaluation, consistent with both retaining alignment and uniformity properties that $\mathcal{L}_{\text{NCE}}$ indirectly optimizes.

2.4.3 Representational Geometry

Representational geometry describes the global structure of the learned feature space. Its purpose here is diagnostic: a representation that has collapsed to a low-rank subspace cannot be interpreted meaningfully, so geometry metrics establish whether results from the interpretability and sparsity sections can be trusted. Alignment and uniformity are used in [Section 3](#) to characterize the failure mode that bounds the usable temperature range of NNCL. Effective rank is tracked across all three experiments as a continuous indicator of the same failure mode.

2.4.3.1 Alignment and Uniformity

Wang and Isola [3] formalize the two properties that a good contrastive representation should satisfy. Given a distribution p_{pos} over positive pairs and a unit-norm encoder f , alignment measures how close the representations of positive pairs are:

$$\mathcal{L}_{\text{align}}(f) = \mathbb{E}_{(x,y) \sim p_{\text{pos}}} [\|f(x) - f(y)\|_2^2] \quad (18)$$

Because f is unit-norm, expanding the squared distance gives $\|f(x) - f(y)\|_2^2 = 2 - 2f(x)^\top f(y)$, so minimizing alignment loss is equivalent to maximizing the cosine similarity between positive pairs, the same quantity that appears in the numerator of the NT-Xent loss.

Uniformity measures how evenly the full distribution of representations covers the hypersphere, using the log of the mean pairwise Gaussian kernel:

$$\mathcal{L}_{\text{uniform}}(f) = \log \mathbb{E}_{x,y \stackrel{\text{i.i.d.}}{\sim} p_{\text{data}}} [e^{-2\|f(x)-f(y)\|_2^2}] \quad (19)$$

A lower value corresponds to a more uniform distribution and indicates that the encoder is not discarding information by collapsing features onto a small region of the space. Using the same

expansion $\|u - v\|_2^2 = 2 - 2u^\top v$, the Gaussian kernel reduces to $\exp(4f(x)^\top f(y) - 4) \propto \exp(f(x)^\top f(y)/\tau)$, the same form as each term in the NT-Xent denominator. Both metrics are computed on the L2-normalized projector output z , using the same positive pairs as the training objective.

Separating the two losses localizes where a failure occurs: a spike in uniformity with stable alignment indicates collapse in spread, while the reverse indicates positives are drifting apart.

2.4.3.2 Rank and Representation Collapse

To diagnose dimensional collapse [4], the singular value spectrum of the centred feature matrix $Z \in \mathbb{R}^{N \times D}$ is computed. Each singular value measures how much variance the corresponding dimension captures across the dataset: large singular values correspond to directions that are actively used to separate inputs, while near-zero singular values correspond to dimensions that are essentially constant across all samples. The shape of this spectrum is informative in itself: a representation that uses the full dimensional capacity produces a gradually decaying curve, whereas dimensional collapse can appear as a cliff, dropping orders of magnitude within a few steps.

Effective rank [28] aggregates the full spectrum into a single scalar that quantifies how evenly variance is distributed. Rather than hard rank, which is integer-valued and sensitive to noise, it treats the normalized singular values $p_k = \sigma_k / \sum_j \sigma_j$ as a probability distribution and takes the exponential of their Shannon entropy:

$$\text{erank}(Z) = \exp\left(-\sum_k p_k \log p_k\right) \quad (20)$$

A value near D indicates that variance is spread evenly across all dimensions; a value near 1 indicates near-total collapse onto a single direction.

Effective rank and the singular value spectrum measure how much of the encoder’s dimensional capacity is actually in use. A representation that only uses a lower-dimensional subspace will have a low effective rank and a sharp drop-off in singular values.

2.4.4 Downstream Performance

A linear classifier is trained on top of the frozen encoder to predict object identity and category on RT4K, following the standard linear probing protocol [1], [29]. Both the backbone representation h and the projector output z are probed.

The two probes differ in what they hold out, following Section 2.2.1. The identity probe covers the 512 training objects and their held-out odd-indexed views. The category probe uses the objects outside that subset.

Probing measures how much information is linearly decodable from the representation at all, independently of whether that information is concentrated in individual units. It serves as a ceiling for neuron-level matching: if the linear probe cannot decode a concept, no single-unit metric will either.

3 Characterizing the Non-Negativity Gate

This chapter characterizes how the contrastive temperature τ influences the representations learned with and without the non-negativity gate.

Two training variants are compared: standard contrastive learning (CL) and Non-Negative Contrastive Learning (NNCL) [6], which adds the non-negativity gate described in [Section 2.3.4](#). The temperature is swept over a broad range $\tau \in [0.02, 0.5]$, and both projector configurations of [Section 2.3.2](#) are trained at every temperature. No explicit sparsity regularization is applied ($\lambda = 0$); the interaction between the non-negativity gate and the sparsity regularization of [Section 2.3.5](#) is the subject of [Section 4](#).

All experiments in this chapter use the random pair sampling strategy of [Section 2.2.1](#): the two views of an object are drawn at random from its rotation range, so the angular distance within a positive pair is unconstrained. No pair gap is involved.

The chapter answers five questions in turn:

1. Does the gate achieve sparsity by permanently silencing units, and if so, how severely? ([Section 3.1](#))
2. Is training stable under the non-negativity gate, and where does it break down? ([Section 3.2](#))
3. What sparsity does the gate produce in the stable regime, and what does it reflect? ([Section 3.3](#))
4. Does that sparsity translate into more interpretable individual units and more semantically structured representations? ([Section 3.4](#), [Section 3.5](#))
5. Does the gate trade off representational quality for interpretability, or do the two coexist? ([Section 3.6](#))

Three conditions have to hold before the central question can be answered. The gate must not achieve its sparsity merely by silencing units permanently, since that would conflate lost capacity with selectivity. It must remain stable across a meaningful temperature range, since collapse forecloses interpretation entirely. And the sparsity it produces in the stable regime must reflect genuine selective coding rather than an artefact of the objective. Only when all three hold does it make sense to ask whether the sparsity translates into unit-level semantic structure, which is the fourth question and the one the chapter is ultimately after. The fifth follows as a practical corollary: interpretability that comes at the cost of downstream usability is of limited value.

The chapter also fixes two parameters for all subsequent experiments: the projector configuration (settled in [Section 3.1](#)) and the operating temperature τ (settled in [Section 3.5](#)).

3.1 Projector Configuration and Dead Neurons

The non-negativity gate can silence units permanently [30]. Under the contrastive objective, units pushed to always-negative pre-activations stop receiving gradient ([Section 2.3.4](#)) and cannot recover. A model with many such dead units appears highly sparse, but the sparsity reflects lost capacity rather than selective coding.

As described in [Section 2.3.4](#), Wang et al. [6] address this with a straight-through estimator that routes gradients through GELU during the backward pass, keeping dead units recoverable.

The figure below shows the two activation functions; GELU’s non-zero derivative for mildly negative inputs is what allows recovery.

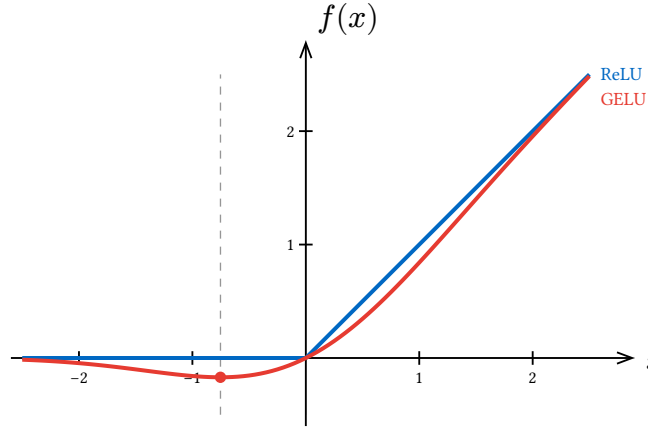


Figure 3: ReLU and GELU. NNCL keeps the ReLU forward pass but routes gradients through GELU, whose derivative is smooth and positive for mildly negative inputs ($x \gtrapprox -0.752$).

The optional batch normalization layer in [Section 2.3.2](#), placed right before the gate, addresses this differently. Units are centred and scaled to match a learned mean and variance:

$$\hat{z} = \gamma \frac{\tilde{z} - \mu}{\sigma} + \beta \quad (21)$$

where μ and σ are the unit’s batch statistics and γ, β are learned scale and shift parameters per unit. The standardized score $s = (\tilde{z} - \mu)/\sigma$ has zero mean and unit variance over the batch, so the unit activates exactly when $s > -\beta/\gamma$.

This provides a direct lever to control each unit’s level of sparsity: in this setup, the uniformity pressure on the representations pushes most units towards deactivation. The learned bias β and scale γ shift and scale the distribution into the negative range, so that only a small tail remains positive and active.

Without batch norm, no such lever exists. The firing rate of a unit depends on where its pre-activation distribution sits in absolute terms, which is set by the projector weights and the incoming feature statistics, both of which drift during training. Sparsity is then a byproduct of where that distribution happens to sit rather than a quantity the objective can set.

The fixed scale also changes what the straight-through estimator sees. Since the standardized score keeps unit variance, a unit held at a low firing rate by γ and β is likely to sit just below the gate, where it can be influenced by the positive GELU gradient.

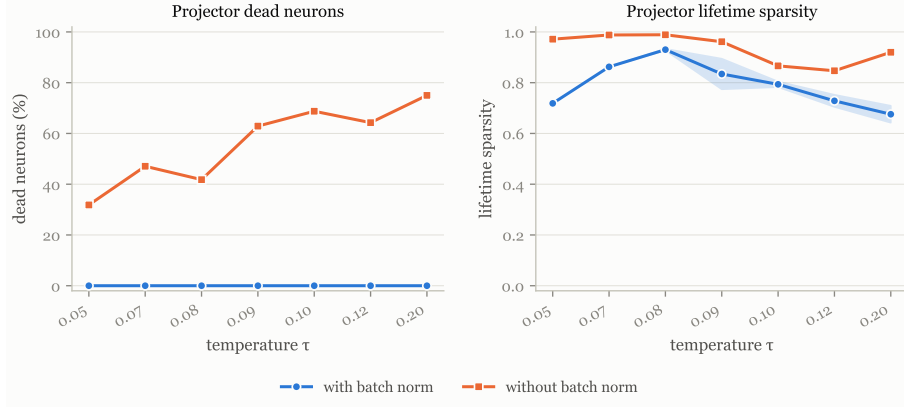


Figure 4: Dead neurons and lifetime sparsity across temperature τ , with and without batch normalization. Left: percentage of dead neurons; right: average lifetime sparsity.

With batch normalization the percentage of dead neurons falls to zero across the entire temperature range, while without it a substantial share of units is permanently inactive. Average lifetime sparsity is higher without batch normalization, but only because dead units count as maximally sparse and inflate the measure. Since the difference reflects lost capacity rather than more selective coding, batch normalization is used in the projector for all following experiments.

3.2 Stability and Critical Temperature

The non-negativity gate constrains representations to the positive orthant, which may interact with the contrastive objective in ways that destabilize training. Whether this interaction is benign across the full temperature range, and what happens when it is not, is the subject of this section.

3.2.1 Representational Geometry

Alignment and uniformity [3] decompose the contrastive objective into two geometric properties, providing insights into how well the resulting representation satisfies the requirements of the contrastive loss.

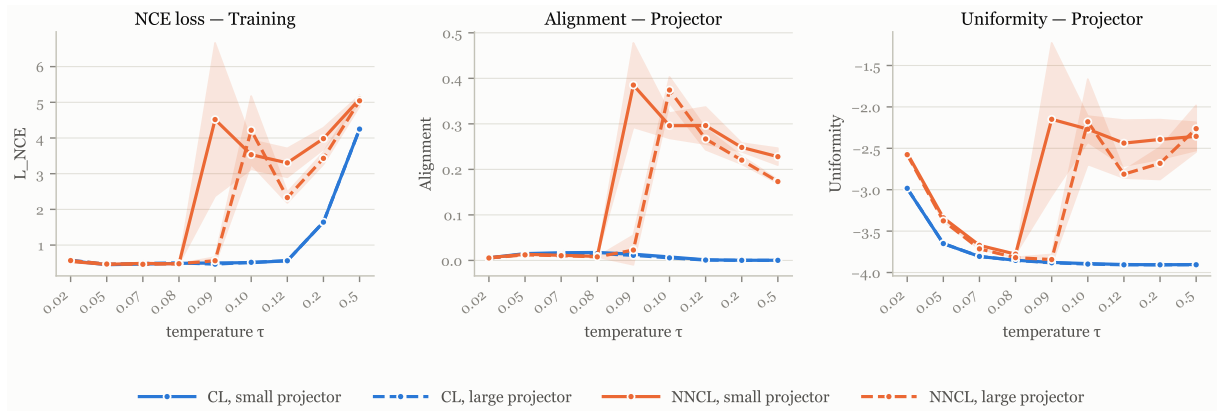


Figure 5: Training and representation losses at the projector across temperature τ . Left: NT-Xent training objective $\mathcal{L}_{\text{NCE}}$; middle: alignment loss $\mathcal{L}_{\text{align}}$; right: uniformity loss $\mathcal{L}_{\text{uniform}}$.

For **CL**, all three losses remain stable across the full temperature range. For **NNCL**, all losses spike sharply at $\tau \approx 0.1$, with $\mathcal{L}_{\text{align}}$ partially improving at higher temperatures while $\mathcal{L}_{\text{uniform}}$ remains elevated.

This indicates that at $\tau \approx 0.1$, **NNCL** converges to a degenerate solution: the non-negativity constraint interacts with the temperature in a way that destabilizes training. The full singular value spectra identify which degenerate solution it is.

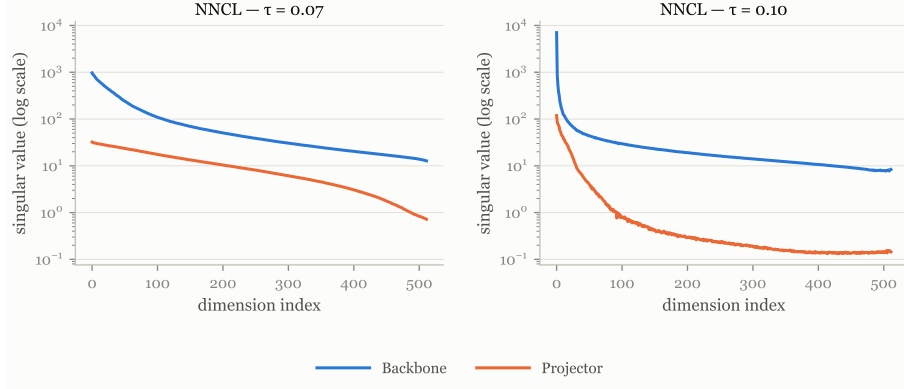


Figure 6: Singular value spectra of **NNCL** representations at $\tau = 0.07$ (left) and $\tau = 0.10$ (right). Backbone and projector spectra are shown separately; dimensions are sorted by singular value (log scale).

At $\tau = 0.07$, singular values decay at a smooth exponential rate across both backbone and projector. At $\tau = 0.10$, the leading singular values are disproportionately large and fall off sharply, with the projector showing a long tail of near-zero values. The loss spike and this rank drop together identify the failure as dimensional collapse [4].

3.2.2 The Margin Bound and Gradient Pressure

The collapse and the sparsity are not separate phenomena: both are outcomes of the same gradient pressure at different intensities. Understanding the mechanism is what justifies temperature as the natural parameter to control sparsity, and explains why the operating regime is not arbitrary.

The NT-Xent loss (Section 1.1.2) is the negative log of the softmax probability p_{ij} that an anchor i assigns to its positive j against all negatives in the batch, $\ell(i, j) = -\log p_{ij}$, with

$$p_{ij} = \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{l=1}^{2N} \mathbb{1}_{[l \neq i]} \exp(\text{sim}(z_i, z_l)/\tau)}. \quad (22)$$

Rewriting $s_{ij} = \text{sim}(z_i, z_j)$ and factoring $\exp(s_{ij}/\tau)$ gives:

$$p_{ij} = \frac{\exp(s_{ij}/\tau)}{\sum_{l \neq i} \exp(s_{il}/\tau)} = \frac{\exp(s_{ij}/\tau) / \exp(s_{ij}/\tau)}{\sum_{l \neq i} \exp(s_{il}/\tau) / \exp(s_{ij}/\tau)} = \frac{1}{\sum_{l \neq i} \exp((s_{il} - s_{ij})/\tau)} \quad (23)$$

The softmax probability p_{ij} depends only on the scaled similarity differences $(s_{ij} - s_{il})/\tau$, where $s_{ij} - s_{il}$ is the margin the positive holds over negative l .

The achievable margin, however, is bounded well below the nominal range of the cosine similarity. Perfect alignment gives at most $s_{ij} = 1$, and the negatives cannot all be pushed to the far side of the sphere at once: even in the maximally separated configuration, the mean pairwise cosine similarity of many points on a unit sphere lies only marginally below zero [31]. Near-orthogonality is therefore effectively the best that can be reached on average, and the margin the positive can hold over a negative is approximately 1.

Assuming every negative holds the same margin $g = s_{ij} - s_{il}$, the sum then collapses: the term $l = j$ is the positive itself and contributes $\exp(0) = 1$, while each of the remaining $m = 2N - 2$ negatives contributes $\exp(-g/\tau)$:

$$p_{ij} = \frac{1}{\sum_{l \neq i} \exp((s_{il} - s_{ij})/\tau)} \approx \frac{1}{1 + m \exp(-g/\tau)} \quad (24)$$

Define the *ideal encoder* as one that achieves perfect alignment of positives and near-orthogonal, uniformly spread negatives, giving $g \approx 1$. It attains

$$p_{\text{ideal}}(\tau) := \frac{1}{1 + m \exp(-1/\tau)}. \quad (25)$$

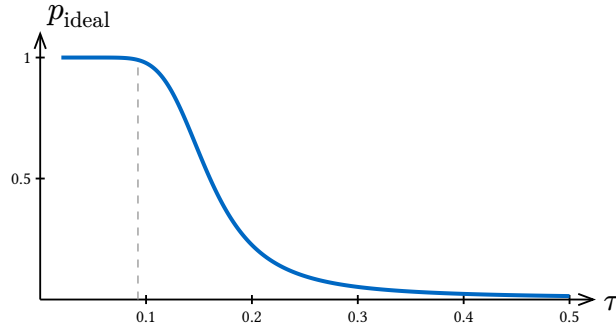


Figure 7: The softmax probability $p_{\text{ideal}}(\tau)$ attained by an ideal encoder, for $m = 510$ negatives (batch size 256). The dashed line marks $\tau = 0.092$, where p_{ideal} falls below 0.99.

The curve stays almost fully saturated up to $\tau \approx 0.09$ and drops sharply afterwards. This threshold coincides with the observed dimensional collapse for **NNCL** variants (Section 3.2.1).

Training minimizes $\ell(i, j) = -\log p_{ij}$, equivalently it maximizes $\log p_{ij}$, so the gradient pressure on the margin g is

$$\mathcal{P}(g, \tau) := \frac{\partial \log p_{ij}}{\partial g} = \frac{1 - p_{ij}}{\tau} = \frac{m \exp(-g/\tau)}{\tau(1 + m \exp(-g/\tau))} \quad (26)$$

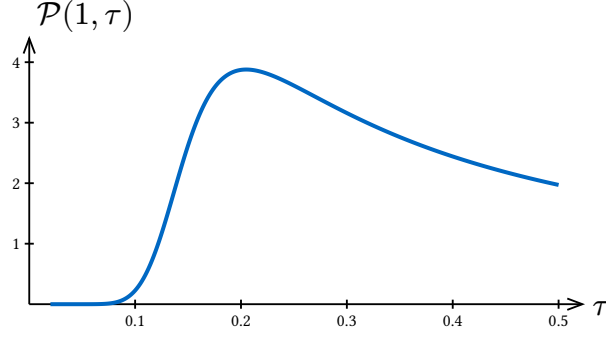


Figure 8: The gradient $\mathcal{P}(1, \tau)$ of the objective with respect to the margin, at the ideal encoder, for $m = 510$ negatives (batch size 256).

In the low-temperature regime, the gradient acting on an ideal encoder is close to zero. After about $\tau = 0.09$, the pressure for further increasing the margin rises significantly. Practically, fully satisfying this push for more margin is not possible.

Whether trained encoders actually operate near this ideal and how the rising gradient pressure influences the learned geometry can be checked directly:

τ	POSITIVE s_{ij}	NEGATIVE s_{il}	MARGIN g
0.020	0.9972 ± 0.0023	0.0146 ± 0.3343	0.9826 ± 0.3344
0.050	0.9925 ± 0.0054	-0.0006 ± 0.1824	0.9931 ± 0.1825
0.070	0.9915 ± 0.0059	-0.0017 ± 0.1168	0.9932 ± 0.1169
0.080	0.9912 ± 0.0061	-0.0015 ± 0.0885	0.9927 ± 0.0887
0.090	0.9931 ± 0.0048	-0.0024 ± 0.0653	0.9955 ± 0.0654
0.100	0.9962 ± 0.0026	-0.0012 ± 0.0453	0.9974 ± 0.0453
0.120	0.9994 ± 0.0004	-0.0012 ± 0.0213	1.0006 ± 0.0213
0.200	0.9999 ± 0.0001	-0.0017 ± 0.0079	1.0016 ± 0.0079

Table 1: Mean cosine similarity over positive and negative pairs at the projector and the margin between them, across temperature τ for **CL** at the small projector.

Average similarities for positives and negatives are close to the theoretical optimum for all temperatures. Two trends can be observed:

1. As temperature rises, the standard deviation in negative similarities and margins decreases significantly. At low temperatures, the softmax score p_{ij} can still be saturated with smaller similarity differences than 1, and there is little pressure to further increase this difference. This allows deviations from the ideal margin without impacting the training loss by much. At higher temperatures, the deviation is small, confirming that trained encoders closely track the ideal encoder and that the theoretical analysis applies to the actual training regime.
2. Average margins increase slightly but consistently with higher temperature. This effect is caused by both higher similarities in positives and lower similarities in negatives. This can

be explained by the gradient pressure $\mathcal{P}$, which grows with temperature and pushes the margin further at every step.

3.2.3 Why the Positive Orthant Cannot Absorb It

In **CL**, this pressure is harmless. The full unit sphere leaves room to push negatives marginally past orthogonality (Table 1). Due to the geometric constraints, the margin can only increase slightly. The solution settles at a lower softmax probability p_{ij} and the resulting representation still achieves decent alignment and uniformity.

In **NNCL**, the representation is constrained to the positive orthant, where cosine similarities between negatives are strictly positive. The margin cannot exceed 1, and reaching even that value requires exact orthogonality between all negative pairs, which in this geometry means that any two samples must not share a single active unit. The bound is both tighter and harder to approach than on the full sphere.

In an attempt to increase the similarity difference further, the optimizer pushes most units towards zero on most samples. The increasing loss gradient with respect to margin explains directly why sparsity rises with temperature in Section 3.3. At the extreme end the representation loses rank, and the form this takes depends on the architecture.

Without batch normalization, a unit can be pushed below the gate on every sample, and once it sits far enough below it is unlikely to recover. Part of the sparsity is then no longer selective deactivation but permanently inactive units, and the number of usable dimensions falls.

With batch normalization (Section 3.1), the units are re-centred in every batch, so permanent deactivation becomes unlikely. The representation instead loses rank with every unit still active: large groups of units activate on the same samples, so each contributes fewer independent directions and the code contracts into a lower-dimensional subspace.

3.2.4 Recovery at Higher Temperature

Interestingly, empirical results show the representation quality dropping most significantly at exactly $\tau = 0.1$ and slightly recovering at even higher temperatures. This can be explained by investigating the representation quality per epoch.

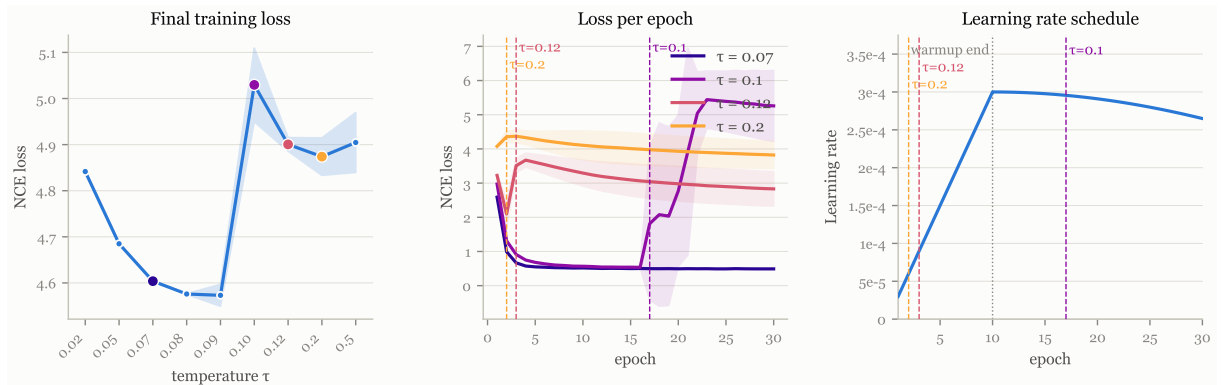


Figure 9: **NNCL** (large projector) training dynamics. Left: converged $\mathcal{L}_{\text{NCE}}$ across temperatures. Middle: $\mathcal{L}_{\text{NCE}}$ per epoch (first 30 epochs) for $\tau \in \{0.07, 0.10, 0.12, 0.20\}$, with the epoch of the loss jump marked. Right: learning rate schedule over the same window.

The converged $\mathcal{L}_{\text{NCE}}$ peaks at $\tau = 0.10$ and is somewhat lower at higher temperatures. Looking at the per-epoch curves, the loss jump shifts earlier as temperature increases: at $\tau = 0.10$ it occurs around epoch 16, whereas at $\tau = 0.12$ and $\tau = 0.20$ it occurs within the first two or three epochs. Notably, jumps that occur later in training are larger in magnitude.

The cosine annealing schedule with warmup provides an explanation. Early in training the learning rate is low; a loss jump at this point causes only a small increase and the model can partially recover. At $\tau = 0.10$, the jump occurs near the end of the warmup phase, when the learning rate is close to its maximum. The resulting loss increase is the largest across all temperatures, which accounts for the worst converged $\mathcal{L}_{\text{NCE}}$. Contrary to what the converged loss alone suggests, higher temperatures appear to be more unstable rather than less, as their loss spikes earlier. All models above $\tau \approx 0.10$ show signs of converging to a low-rank solution, consistent with the dimensional collapse observed in [Section 3.2.1](#).

3.3 Sparsity

Sparsity is measured at the projector output z , where the non-negativity gate applies. Population sparsity and Hoyer sparsity ([Section 2.4.1](#)) are reported here.

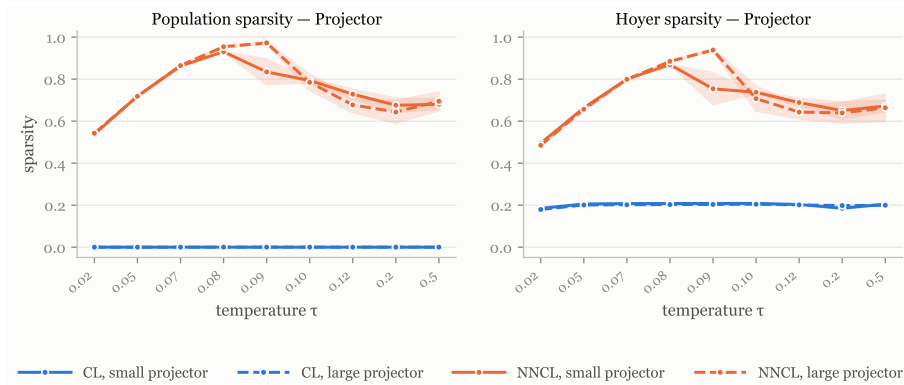


Figure 10: Projector sparsity across temperature τ . Left: population sparsity; right: Hoyer sparsity.

Both sparsity measures increase steadily with temperature, peaking at $\tau = 0.08$. Above this point, sparsity drops and remains lower throughout the high-temperature regime.

The non-negativity constraint restricts representations to the positive orthant. To achieve high uniformity, representations must spread across this constrained space, which pushes them towards the coordinate axes. The non-negativity gate allows turning units off selectively, making the majority of units exactly zero.

In comparison, CL models show no significant sparsity across the full temperature range, as the inducing mechanism is absent in standard contrastive learning.

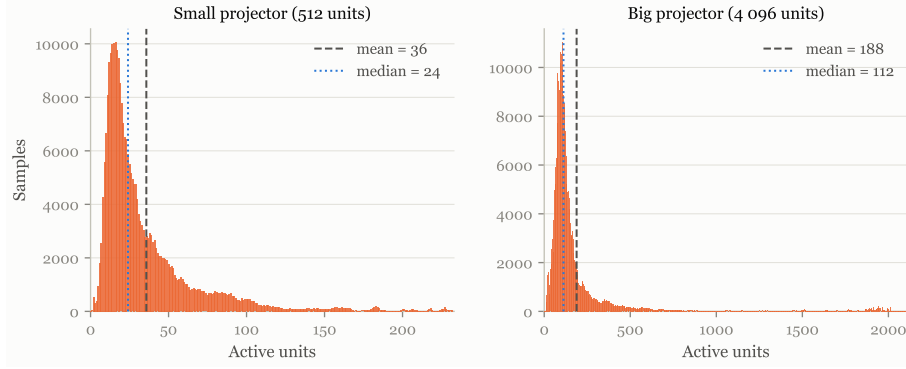


Figure 11: Distribution of absolute active units for **NNCL** at $\tau = 0.08$. Left: small projector; right: large projector.

Both projectors produce strongly right-skewed distributions: most samples activate only a small fraction of the available units, while a minority activate far more. The range is wide in both cases, spanning from a handful of active units at one extreme to close to half of all units at the other. Sparsity is therefore not a uniform property of the representation but varies from sample to sample, with some objects encoded by a compact set of units and others using a much larger portion of the code.

Within a single trained model, each extreme is dominated by one object: the samples at either end are many different views of the same object, spanning the rotation range. Which object that is varies across seeds and projector sizes, though a small set recurs. The number of active units therefore encodes identity in its own right.

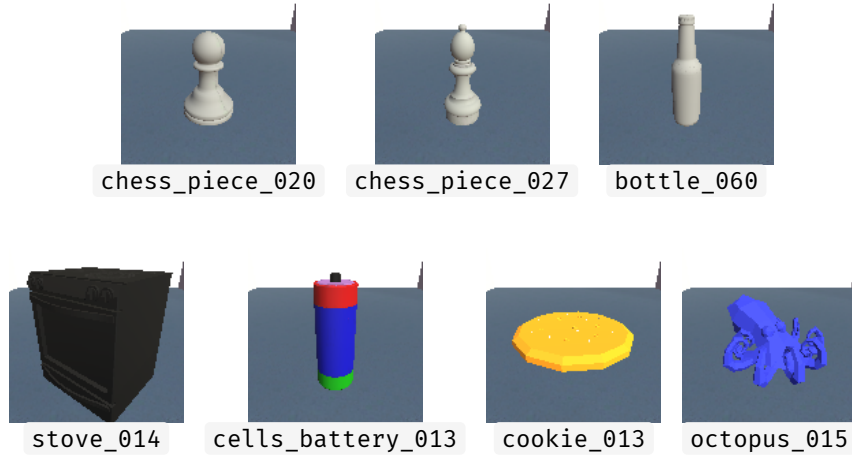


Figure 12: Objects at the extremes of the active-unit distribution for **NNCL** at $\tau = 0.08$. Top: the most sparsely encoded objects. Bottom: the most densely encoded objects, at both projector sizes.

The two groups differ visually. The sparsely encoded objects are small in the frame and pale; the densely encoded ones fill much of the image and stand out strongly against the background. The most extreme case is instructive: on the small projector, one chess piece is encoded by a single active unit across many views. The small projector’s output dimension was chosen to match the number of training objects, with the intent that it might allocate one dimension per identity ([Section 2.3.2](#)); for the simplest objects this is realized literally.

A plausible explanation is that thin objects with few distinguishing visual features need correspondingly few units to describe. This suggests the code is adaptive rather than uniformly sparse: the number of active units tracks the visual complexity of the object, with simpler appearances requiring fewer dimensions to distinguish. That the model arrives at this allocation without any explicit pressure to do so indicates that the sparsity is meaningful, a reflection of content rather than an artefact of the gate. It also points to the units themselves acting as visual feature detectors: an object with little visual structure simply does not trigger many of them.

3.4 Interpretability

Neuron-level matching (Section 2.4.2.1) quantifies how well individual neurons selectively encode object identity and category.

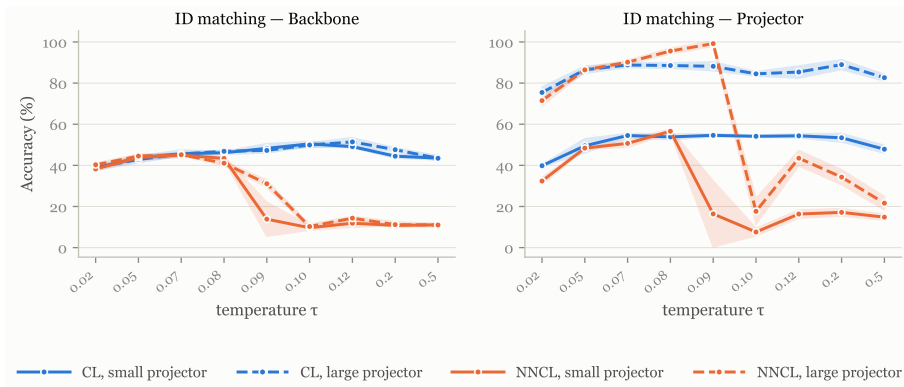


Figure 13: Identity matching across temperature τ . Left: backbone; right: projector.

CL models are stable across all temperatures at both layers, while **NNCL** shows a large drop in representational quality at higher τ , consistent with the dimensional collapse in Section 3.2.1.

At the backbone, matching is unaffected by model variant or projector size (besides the dimensional collapse for **NNCL**), suggesting that neither the projector nor the variant shapes per-unit identity selectivity in earlier layers.

At the projector, the larger variant yields a substantial advantage for both models: the small projector’s 512 units must cover all 512 identities with no room to discard poorly selective neurons, whereas the large projector selects 512 from a pool of 4096 and leaves the rest unmatched.

At the ideal temperature, modest improvements in matching are observed at the projector layer. One likely reason is that higher sparsity concentrates each identity’s representation onto fewer units, requiring each active unit to carry more of the discriminative signal and thereby increasing per-unit selectivity.

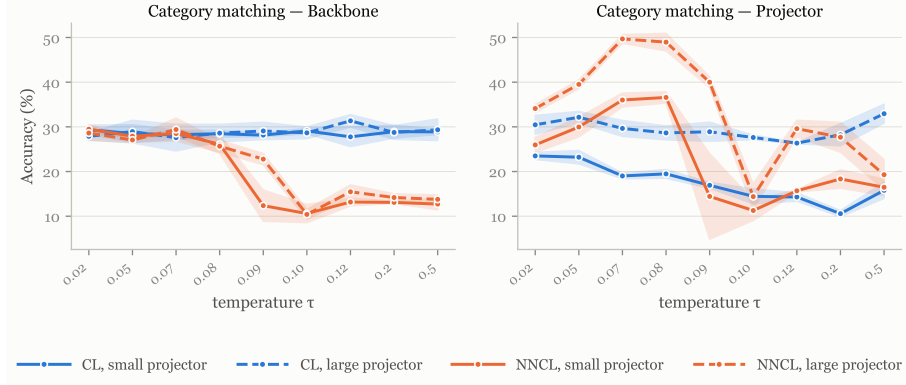


Figure 14: Category matching across temperature τ . Left: backbone; right: projector.

The same temperature instability in **NNCL** is visible for category matching, while **CL** remains stable. The dominant effect is the model variant: **NNCL** outperforms **CL** by a wide margin at the projector layer. The advantage increases with temperature, peaking at $\tau = 0.08$ before collapsing, a shape that is similar to the population sparsity (Section 3.3), indicating the improvement is driven by the selectivity that sparsity produces.

At the projector, a larger projector improves category matching, but the gain is smaller than for identity. With 512 units and only 104 categories, even the small projector offers roughly five candidate units per class, so the additional capacity matters less than it does for identity. The per-object accuracy distribution shows what kind of selectivity **NNCL** actually produces.

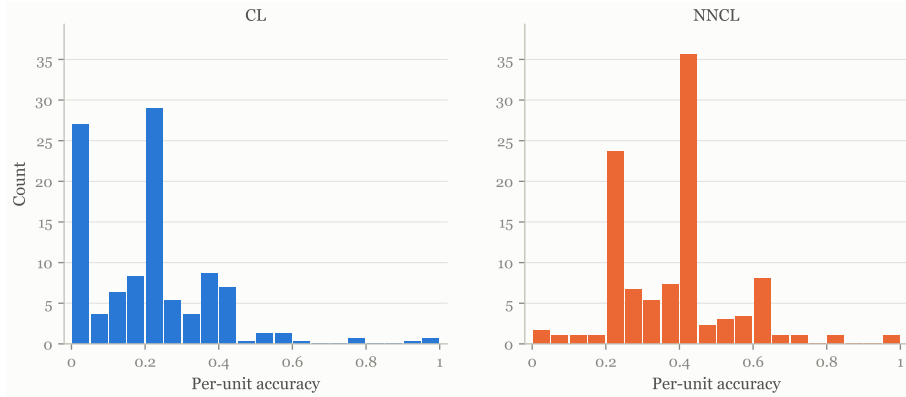


Figure 15: Histogram of per-matched-unit category accuracy at the small projector ($\tau = 0.08$, 104 categories). Each matched unit contributes one accuracy, the fraction of views of its matched category that it wins; bars give the mean number of units per accuracy bin, averaged over 3 seeds. Left: **CL**; right: **NNCL**.

NNCL shifts the per-matched-unit accuracy distribution significantly to the right compared to **CL**, making the matching advantage concrete. The distribution is not smooth: pronounced spikes appear at 0%, 20%, 40%, and 60%. These discrete steps follow from the metric and the dataset structure. A view counts as correct when its category’s matched unit wins the argmax over all 104 matched units (Section 2.4.2.1). The training objective pulls views of an object together, so a unit that wins on one view is likely to win all. Accuracy then moves in steps of one object rather than one view, and with roughly five objects per category, a unit that wins on two of them scores 40%, on three 60%. The mass between the spikes comes from objects the unit wins only partially, where invariance is imperfect and some views of the object fall to a competing unit.

The spikes are not equivalent. The 0% bin is a consequence of the one-to-one assignment, which assigns every category a unit whether a suitable one exists or not. Winning on exactly one object is what a purely identity-selective unit scores when asked a category question, so the 20% spike is a baseline rather than evidence of category structure. The evidence is the mass above it: a unit at 40% or 60% wins on two or three distinct objects that share a category, even though training treats those objects as negatives and pushes them apart.

This is evidence that category-selective units emerge from training, without any direct learning signal for category membership. This emergent semantic structure arises purely from the sparsity constraint interacting with the contrastive objective, showing the model self-organizing its code around visual properties shared across objects within a category.

3.5 Disentanglement

Based on previous results, $\tau = 0.08$ is selected and held fixed for evaluating disentanglement. It lies within the stable regime of **NNCL** and avoids the representational collapse observed at $\tau \geq 0.1$. Across all evaluated metrics, representational quality at $\tau = 0.08$ is either optimal or within a negligible margin of the best observed value. Fixing this parameter drastically reduces the complexity for further analysis.

SEPIN (Section 2.4.2.4) measures whether the structural constraints of **NNCL** translate into more disentangled representations.

	SEPIN@1	SEPIN@10	SEPIN@100	SEPIN@ALL
CL, small	2.4 ± 0.0	2.3 ± 0.0	2.1 ± 0.0	1.9 ± 0.0
CL, large	0.3 ± 0.0	0.3 ± 0.0	0.3 ± 0.0	0.2 ± 0.0
NNCL, small	3.7 ± 0.3	3.3 ± 0.1	2.7 ± 0.0	1.9 ± 0.0
NNCL, large	1.5 ± 0.1	1.3 ± 0.0	0.9 ± 0.0	0.2 ± 0.0

Table 2: SEPIN of projector representations at $\tau = 0.08$, mean $\pm$ std over 3 seeds. Values scaled by $\times 10^3$.

The projector representation is clearly more disentangled under **NNCL**, with higher scores across all tested subsets of dimensions ($k \in \{1, 10, 100\}$) for both projector variations. This replicates the finding of Wang et al. [6] in a controlled 3D object recognition setting.

Wang et al. [6] explain this advantage with the breaking of rotational symmetry. The $\mathcal{L}_{\text{NCE}}$ loss depends on the features only through pairwise similarities, which makes it invariant to rotations of the feature space: any rotation of a solution is equally good. There is no incentive to encode meaning along the coordinate axes. Non-negativity removes this invariance, since rotating a non-negative representation introduces negative entries. Axes become meaningful directions rather than an arbitrary choice of basis.

At the same time, uniformity pressure inside the positive orthant pushes representations towards the coordinate axes (Section 3.3). Units switch off for most samples and respond to only a narrow set, which makes them selective. Selectivity is what SEPIN rewards, since it measures the information a single dimension contributes individually. In **CL**, the same infor-

mation is spread over many units at once, so the remaining dimensions largely compensate for any single one and its unique contribution stays small.

	SEPIN@1	SEPIN@10	SEPIN@100	SEPIN@ALL
CL, small	14.9 ± 4.5	8.0 ± 2.9	2.6 ± 1.5	0.0 ± 0.2
CL, large	3.6 ± 5.0	2.7 ± 3.8	0.8 ± 1.2	0.0 ± 0.0
NNCL, small	62.4 ± 6.6	41.3 ± 2.6	16.8 ± 1.4	-2.0 ± 0.5
NNCL, large	71.1 ± 10.1	44.0 ± 11.4	16.2 ± 1.9	-1.1 ± 0.6

Table 3: SEPIN of backbone representations at $\tau = 0.08$, mean $\pm$ std over 3 seeds. Values scaled by $\times 10^3$.

NNCL shows similar advantages in the backbone representation, suggesting that gradients flowing through the non-negativity gate encourage earlier layers to adopt a more disentangled structure. How exactly this effect propagates is not fully understood, and further analysis of this mechanism is beyond the scope of this thesis. Note that absolute SEPIN values are not directly comparable between the projector and backbone representations: backbone features are not normalized to the unit sphere and are z-score standardized before evaluation, placing them on a different scale.

3.6 Downstream Performance

The preceding sections establish that **NNCL** produces sparser, more disentangled representations in the stable regime. The question is whether this comes at a cost to representational quality, or whether interpretability and capacity coexist.

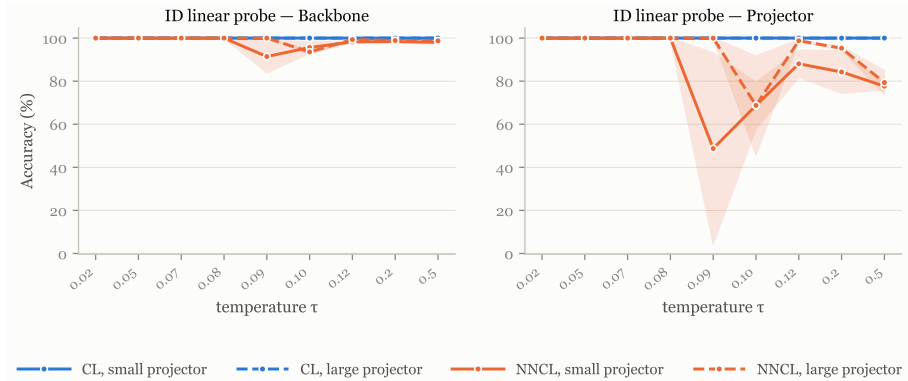


Figure 16: Identity probe accuracy at the backbone (left) and projector (right) across temperature τ .

Both **CL** and **NNCL** achieve near-perfect identity probe accuracy for $\tau < 0.1$. This is expected: identity is directly encoded in the contrastive objective, since all cross-identity pairs are negative examples. The sharp drop at $\tau \approx 0.1$ is an artefact of training collapse, not an inherent cost of the non-negativity gate.

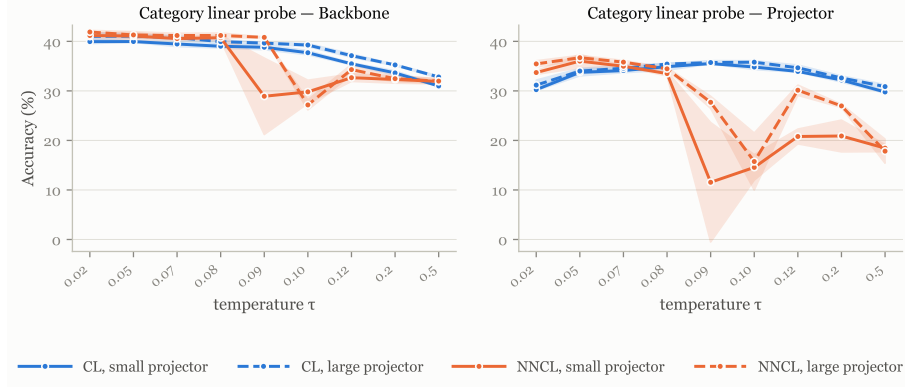


Figure 17: Category linear probe accuracy across temperature τ . Left: backbone; right: projector.

Category probe accuracy tells the same story: **CL** and **NNCL** are comparable across the stable temperature range, with **NNCL** showing a small but consistent advantage at lower temperatures, in line with the findings of Wang et al. [6].

Within the stable regime, the non-negativity gate does not trade off downstream performance for interpretability. The disentanglement and sparsity advantages documented in [Section 3.4](#) and [Section 3.5](#) come without a measurable loss in representational capacity.

4 Explicit Sparsity Regularization

The non-negativity gate enforces sparsity implicitly by zeroing negative activations. This chapter tests an alternative: the explicit sparsity penalty of [Section 2.3.5](#), applied with and without the gate. The penalty is applied to both variants: **CL** is trained with the sparsity term and no non-negativity gate; **NNCL** with both.

This experiment addresses two related questions:

1. Can λ serve as a reliable lever for controlling the degree of sparsity, independently of the contrastive objective?
2. Is axis-aligned sparsity without the structural constraint of non-negativity sufficient to improve interpretability?

The regularization coefficient λ is swept across a range of values. All model configurations from [Section 3](#) are used, with temperature fixed at $\tau = 0.08$.

4.1 Sparsity Control

Projector sparsity is measured across the λ sweep to test whether the penalty provides a usable control lever and how far it can be pushed before the representation degrades.

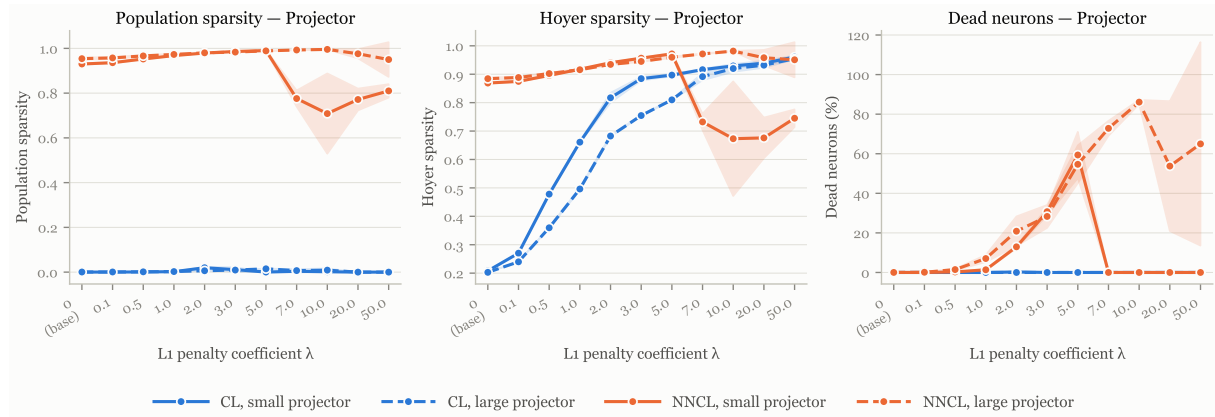


Figure 18: Projector sparsity across regularization coefficient λ . Left: population sparsity; middle: Hoyer sparsity; right: dead-neuron percentage.

With increasing λ , population sparsity in **NNCL** rises steadily, peaking above 99% for both projectors: at $\lambda = 5$ for the small projector and $\lambda = 10$ for the large one. In absolute terms, this leaves only a handful of active units per sample, around 3 of 512 and 4 of 4096 respectively. Beyond these coefficients, sparsity drops sharply, suggesting training collapse.

CL shows no population sparsity across the full λ range, as unit deactivation requires the non-negativity gate. Hoyer sparsity, however, increases for both variants: **CL** starts near zero but rises quickly, matching **NNCL** at higher λ .

The dead-neuron percentage rises in **NNCL**, reaching 60% and 86% at the respective sparsity optima, despite output batch normalization, which would ordinarily prevent this by mean-centring each dimension and pulling dead units back into an active range.

The penalty is a reliable lever for sparsity in both variants. Importantly, **CL** reaches Hoyer sparsity comparable to **NNCL** without deactivating any units. The non-negativity gate is not

required for axis alignment: the same concentration of activation mass onto a small number of dimensions can be achieved through many near-zero activations rather than hard zeros.

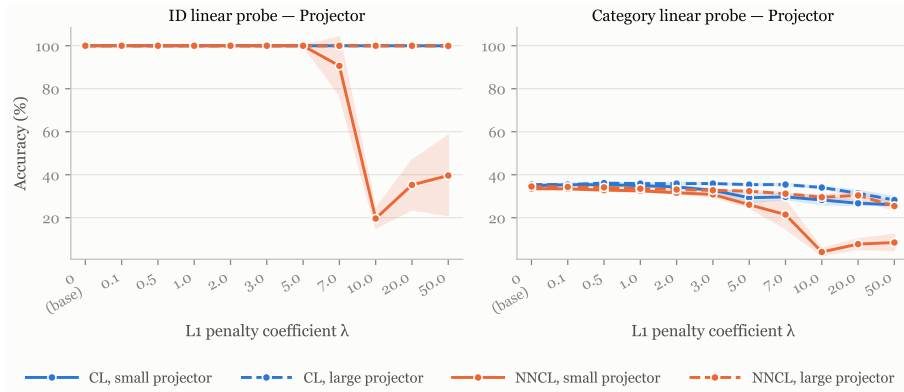


Figure 19: Category and identity probe accuracy at the projector across λ , showing representational collapse in NNCL at high regularization coefficients.

Linear probing accuracy is unaffected by the sparsity regularization across the full λ range, with one exception: **NNCL** with the small projector collapses sharply at $\lambda > 5.0$. Sparsity pressure up to this threshold therefore comes at no cost to representational quality relative to the **CL** baseline. The collapse point marks the upper bound of the usable λ range for **NNCL** (small projector), and is consistent with the rank drop observed in Figure 20.



Figure 20: Effective rank of projector representations across λ . Collapse in **NNCL** at high regularization coefficients is reflected in a sharp rank reduction.

NNCL maintains a flat effective rank across the full range before collapsing sharply: the small projector drops at $\lambda > 5$, the large projector at $\lambda > 10$. **CL**, by contrast, shows a gradual rank increase under regularization, reaching a peak before declining again. The small projector peaks at $\lambda = 1$ and the large projector at $\lambda = 5$, suggesting that moderate sparsity regularization broadens the representational geometry of **CL** up to an optimum, beyond which over-regularization begins to compress it.

The rank increase for **CL** is a first sign that axis-aligned representations benefit representational quality, even without deactivated units. Both sparsity measures are fractions of the projector width, while the units available to carry information are an absolute count. Hoyer sparsity normalizes by the square root of the projector width, so at equal measured sparsity, the large projector still holds eight times as many effectively active units. It therefore absorbs more pressure before the rank drops.

Beyond the respective optima, sparsity and effective rank drop sharply, and the linear probes follow. Maximizing S_H pushes activations towards the coordinate axes, which in the positive orthant are the extreme points of the representation space. The increasing pressure destabilizes training and causes dimensional collapse, similar to [Section 3.2](#). In **NNCL**, the drop is partly explained by dead neurons, despite the batch norm recentring mechanism in [Section 3.1](#). **CL** has no dead units, so the decline is not structural: an excessive penalty competes with the contrastive objective, trading representational spread for sparsity until the two goals are no longer jointly satisfied.

4.2 Interpretability

Interpretability is measured as in [Section 3.4](#) and [Section 3.5](#): unit-level selectivity for semantic concepts, and the degree to which separate dimensions carry independent information.

4.2.1 Neuron Selectivity

Neuron-level matching ([Section 2.4.2.1](#)) shows how selectively individual units activate for measurable semantic concepts (object identity and category).

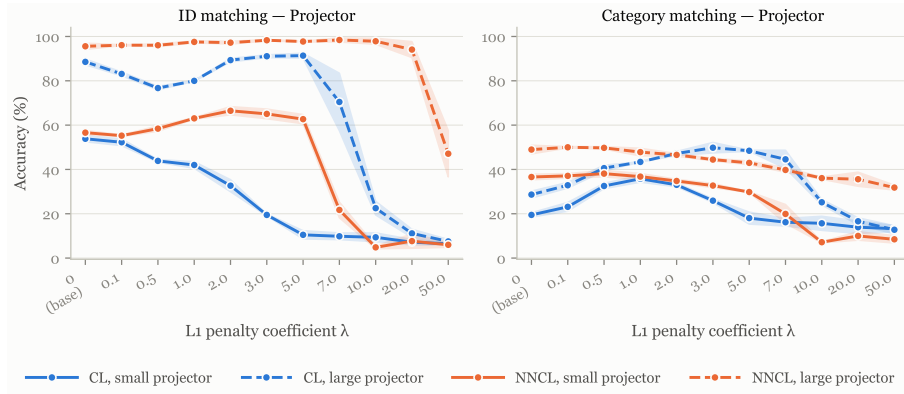


Figure 21: Neuron-level matching at the projector across λ . Top row: identity matching; bottom row: category matching.

At the projector, identity and category matching respond differently to λ . Category matching improves significantly under standard **CL** and matches the **NNCL** baseline. The optimum, however, depends on the projector configuration: the small projector peaks at $\lambda = 1.0$ and drops afterwards. The larger projector performs best at $\lambda = 3.0$. **NNCL** variants peak at $\lambda = 0$, without any additional regularization.

Identity matching shows a more complex pattern. The large **NNCL** projector is already close to perfect and improves further under higher λ . The smaller projector also improves, but drops sharply beyond $\lambda = 5.0$, coinciding with the representational collapse (see [Section 4.1](#)). **CL** loses accuracy steadily under higher λ in the smaller projector configuration. The larger projector shows a bimodal pattern, with peaks at $\lambda = 0$ and $\lambda = 5$, and a moderate dip between them. One plausible explanation is that the penalty disrupts the distributed code that initially supports identity matching before axis-alignment is strong enough to replace it with selective units. This pattern is not further investigated here.

The improvement in category matching for **CL** is particularly significant. Category matching is precisely the metric that rises with sparsity under the non-negativity gate ([Section 3.4](#)),

where it was identified as the clearest signal of emerging semantic structure. The regularization term breaks rotational invariance (Section 1.1.3) through axis-alignment pressure rather than the non-negativity constraint, and the same unit-level semantic structure improves. The finding reinforces the connection between the breaking of rotational invariance and interpretability: the mechanism matters less than the symmetry it removes.

For **NNCL**, rotational invariance is already broken by the gate, so additional axis-alignment pressure from the regularization term does not add to the mechanism but competes with the contrastive objective. Category matching decreases consistently at higher λ . Adding the regularization term on top of the gate actively hurts the structure that the gate already produces.

4.2.2 Disentanglement

Section 3.5 established that the non-negativity constraint produces substantially more disentangled representations. Only **CL** is reported across the λ sweep here, since the open question is whether the regularization term alone produces comparable disentanglement. The reference point is the **NNCL** baseline at $\lambda = 0$ in Table 2, measured at the same temperature and projector configurations.

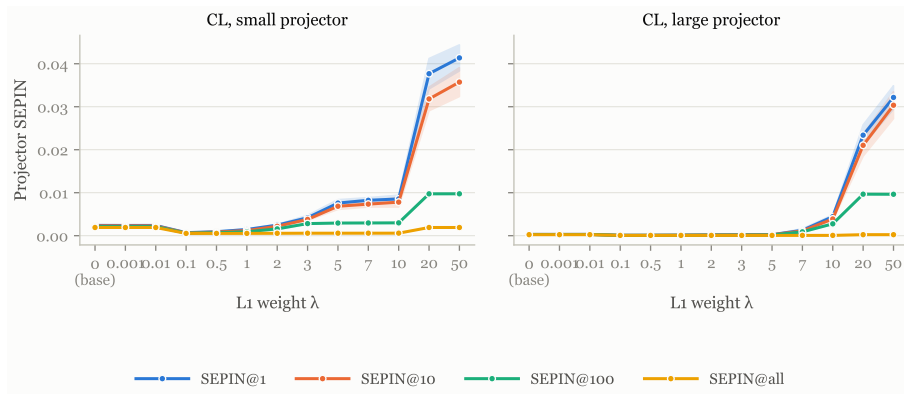


Figure 22: SEPIN of projector representations for **CL** across λ .

Projector disentanglement increases with λ ; SEPIN@1 and SEPIN@10 show the sharpest gains as axis-alignment takes hold, with the largest jumps at the high end of the sweep. Without the non-negativity gate, the sparsity regularization alone increases disentanglement: as activation mass concentrates on fewer dimensions, only those dimensions carry the information needed to separate negatives, and the representation becomes more axis-separated.

4.2.3 Sparsity Without Non-Negativity

Taken together, the two measures answer the second question posed at the start of the chapter: axis-aligned sparsity without non-negativity is sufficient to improve interpretability. Under the sparsity penalty alone, **CL** reaches category matching on par with the **NNCL** baseline, and its projector disentanglement rises well above unregularized **CL** towards that baseline, all without a single deactivated unit. Sparsity in the sense of concentrated activation mass, rather than hard zeros, is therefore what drives the interpretability gains, and the non-negativity gate is one way of obtaining it rather than a requirement.

The two mechanisms are not equivalent, and each retains distinct advantages. The non-negativity gate is the only one that produces hard sparsity: units are fully deactivated, so

every feature admits a binary read as either active or inactive for a given sample, and no hyperparameter has to be tuned to obtain it. Its advantages are also directly measurable in identity matching and effective rank, where **NNCL** stays ahead. These advantages come at the cost of dead neurons and the stability issues established in [Section 3.2](#). The explicit penalty deactivates no units and therefore offers no such binary reading, but it keeps training stable over a wider range and turns sparsity into an adjustable quantity. The coefficient is an additional hyperparameter whose optimum depends on the projector width; it is, at the same time, the only lever available for setting the degree of sparsity directly.

5 Generalization to a Temporal Setting

The previous chapters show that sparsity produces unit-level structure in contrastive representations on RT4K (Section 2.1). The open question is whether that link survives when the positive signal is temporal and the generating factors are continuous rather than discrete, and what form the structure takes if it does. This mirrors the setting in which sparse coding is thought to emerge in biological vision (Section 1.1.5).

The chapter answers it by applying the same comparison of **CL** and **NNCL** to MIMo-Mirror (Section 2.2.2). Two features of the dataset change the measurement problem. Positive pairs are formed from temporal continuity (Section 2.2.2), with the pair gap $g \in \{1, 2, 3\}$ setting how many frames apart each pair is drawn. There are no discrete labels, so selectivity is measured against the continuous ground factors described in Section 2.2.2. Temperatures in the range $\tau \in [0.02, 2]$ are explored. Only the *small projector* of Section 2.3.2 is used: MIMo-Mirror has no discrete object identities and its ground factors are only 8-dimensional, so the additional capacity of the large projector carries little benefit. No explicit sparsity regularization is applied ($\lambda = 0$). Training otherwise follows Section 2.3.

Two questions structure the chapter:

1. Does the conditional instability found in Section 3.2 carry over to a temporal learning setting, and what governs the stable operating regime? (Section 5.1)
2. Does the link between sparsity and unit-level structure extend to a naturalistic, temporal setting? (Section 5.2)

5.1 Stability

Section 3.2 found the non-negativity gate to be conditionally stable on RT4K: below a critical temperature, **NNCL** trained normally; above it the representation collapsed. To investigate whether this instability carries over to the temporal setting, effective rank and training loss are examined across temperatures and pair gaps.

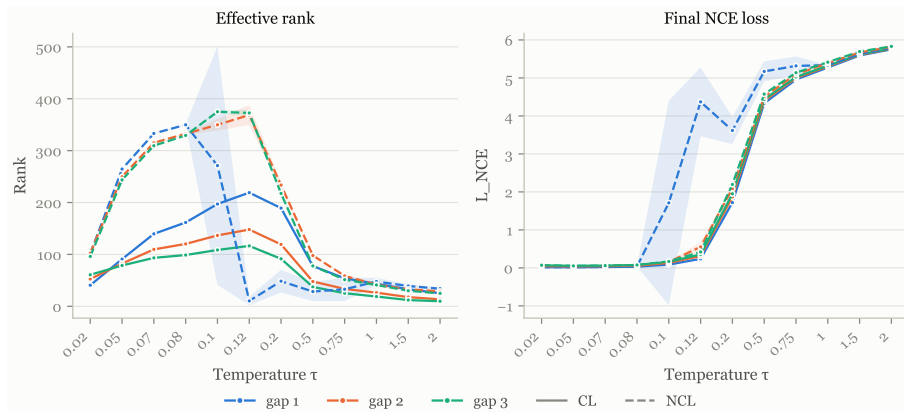


Figure 23: Effective rank and converged training loss $\mathcal{L}_{NCE}$ of the projector across temperature τ per pair gap g . Left: effective rank; right: training loss $\mathcal{L}_{NCE}$.

The **NNCL** $g = 1$ runs stand out: effective rank drops significantly across the temperature range $\tau \in [0.1, 0.75]$, when compared to the other **NNCL** models at higher pair gaps. The same configuration shows increased $\mathcal{L}_{NCE}$ across that temperature range. A rank drop and

a loss spike occurring together are the same symptoms that identified dimensional collapse in [Section 3.2](#). Interestingly, the collapse occurs only at $g = 1$. All configurations with $g > 1$ remain stable across the full temperature range.

At $g = 1$ the two views are consecutive frames of a slow and continuous arm movement, so they are visually similar. The positive is easy to align and the loss is carried mostly by the negative term. The same holds on RT4K, where positives are two random views of one object. Many objects change little between viewpoints, so a large share of the pairs is easy, which is reflected in a mean positive similarity above 0.99 at every temperature ([Table 1](#)). That is the setting in which **NNCL** collapsed. At $g > 1$, in contrast, alignment is harder. A possible reading is that easy positives contribute to the collapse alongside temperature: when the positive is already aligned, the negative term dominates the gradient, so more of the pressure pushes against the margin bound of the positive orthant.

Stability returns above $\tau \approx 0.75$. This is consistent with the shape of $\mathcal{P}(1, \tau)$ in [Figure 8](#), which peaks shortly after the critical temperature and then decays asymptotically towards zero. At $\tau = 2$ the pressure to widen the margin is smaller again, and the run stays stable.

The **CL** effective rank is unimodal across temperature, with a maximum in the intermediate range and lower values at both ends. Three regimes explain the shape:

- At low τ , the softmax score p_{ij} saturates at margins well below the ideal ([Section 3.2](#)), so there is little separating pressure. The representation is free to settle inside a lower-dimensional subspace.
- In the intermediate range, the margin gradient is strong enough to enforce separation while still being satisfiable. Negatives are spread across the available directions and the rank is highest.
- At high τ , the softmax weights over negatives flatten, so every negative is repelled with approximately equal strength, independently of whether it is already close or far. Nearby negatives are therefore no longer penalized significantly harder than distant ones, and close negatives caused by visual similarity are more acceptable. One hypothesis is that this allows the representation to follow the generating factors of the dataset, which span only eight dimensions, and the rank drops accordingly.

5.2 Interpretability

Neuron-level matching ([Section 2.4.2.1](#)), used in [Section 3.4](#) and [Section 3.5](#), requires discrete semantic concepts such as object identity. MIMO’s ground truth is eight continuous joint angles, so two continuous metrics are used instead: per-unit correlation ([Section 2.4.2.2](#)), which tests whether any single unit linearly tracks a joint angle, and activation specificity ([Section 2.4.2.3](#)), which tests whether a unit’s activity alone narrows a joint angle to a small range.

Pair gap $g = 2$ is selected for this comparison, to avoid the dimensional collapse observed in [Section 5.1](#).

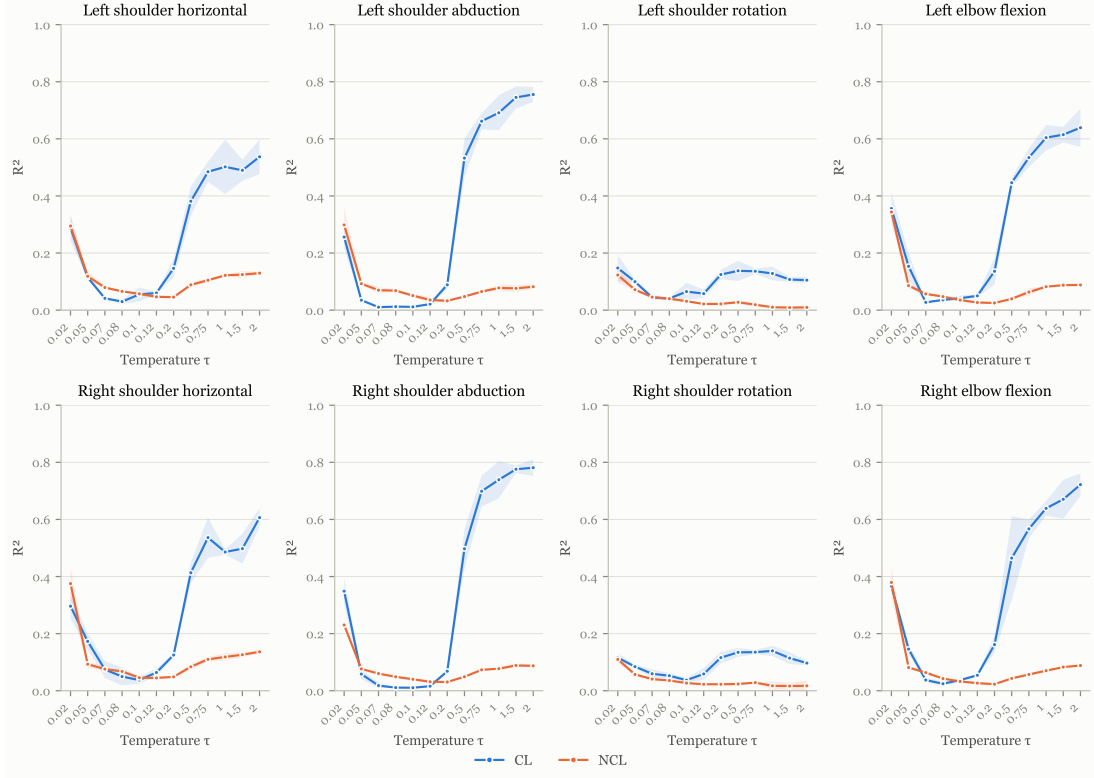


Figure 24: Best per-unit R^2 correlation for each of the eight joint angles, **CL** and **NNCL**.

The curves are nearly symmetric between the left and right arms for every joint, as expected from a movement distribution that treats both arms alike. At the lowest temperature, both variants reach a moderate R^2 between 0.3 and 0.4. In the intermediate range, the best single unit explains almost nothing, between 0.05 and 0.12 for both variants. From $\tau \geq 0.5$ upwards, **CL** rises steeply on every joint, reaching up to $R^2 = 0.78$ (Pearson $r = 0.88$) and exceeding **NNCL** by a wide margin. Shoulder rotation is the exception and stays below 0.15 for both variants across the whole sweep.

The **CL** curve mirrors the inverse of its effective rank (Figure 23): per-unit R^2 is highest exactly where the rank is lowest. This supports the hypothesis stated in Section 5.1: at high temperature, the representation aligns with the subspace of the generating factors, which span only eight dimensions. A representation of low rank is sufficient to encode them. This is consistent with individual units showing high per-unit R^2 : if the representation is confined to a low-dimensional subspace that aligns with the generating factors, single units are more likely to track individual joint angles. This likely depends on the eight generating factors being low-dimensional relative to the projector width of 512.

Shoulder rotation is poorly encoded because rotating the shoulder about its own axis barely changes the image. At the given resolution and with little texture on the arm, the rotation produces almost no visual signal to pick up.

NNCL does not follow **CL** upwards, since its sparsity increases with temperature. A unit that is inactive on most frames cannot track a continuous angle linearly, so per-unit correlation is a poor measure of what a sparse code carries (Section 2.4.2.2).

Activation specificity (Section 2.4.2.3) compares the spread of a joint angle across the whole dataset against its spread restricted to the samples on which a unit is active. Reporting that

restricted spread for every unit separately, instead of reducing it to a single score, shows the shape of the code directly. The densities are shown at $\tau = 2.0$, where unit activation specificity is highest.

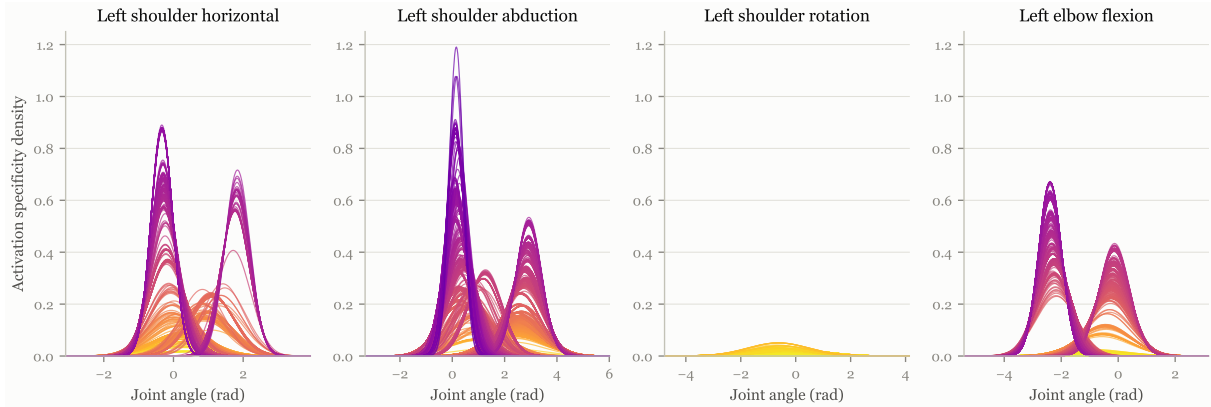


Figure 25: Activation specificity densities over the four left joint angles for NNCL at $\tau = 2.0$ and $g = 2$. Each curve is one unit, drawn as a normal density over the range of joint values on which that unit is active. A tall narrow spike marks a unit that fires only over a small part of the joint range and is therefore highly selective, while a low broad curve marks a unit whose activity says little about the joint. Colour encodes the activation specificity of the unit, running from dark purple at 1 to bright yellow at 0. Units with negative specificity are excluded.

Shoulder rotation again shows no spikes and stays flat, consistent with Figure 24. On the other three joints, two strong spikes emerge at the low and high ends of the angle range, and shoulder horizontal and abduction show a third, smaller spike in the intermediate region. Many units are active over similar ranges rather than covering the angle range evenly between them.

The bimodal shape follows from the dataset. The joint angles are distributed with many frames at the extreme ends and few in the intermediate ranges, so the ranges over which units are active inherit that distribution.

Two reasons could explain why many units cover similar ranges. First, the representation is effectively low-rank at this temperature (Section 5.1), so there are few independent directions to distribute units over. Second, a unit may respond to a full pose rather than to a single joint. A pose is a combination of sub-ranges over all eight angles, and at the low lifetime sparsity observed here, it seems unlikely that a unit is selective for one joint on its own. Projected onto a single joint axis, units would then look alike even when they respond to different poses. This is not tested here.

The same view of the CL representation shows what changes when the non-negativity gate is removed.

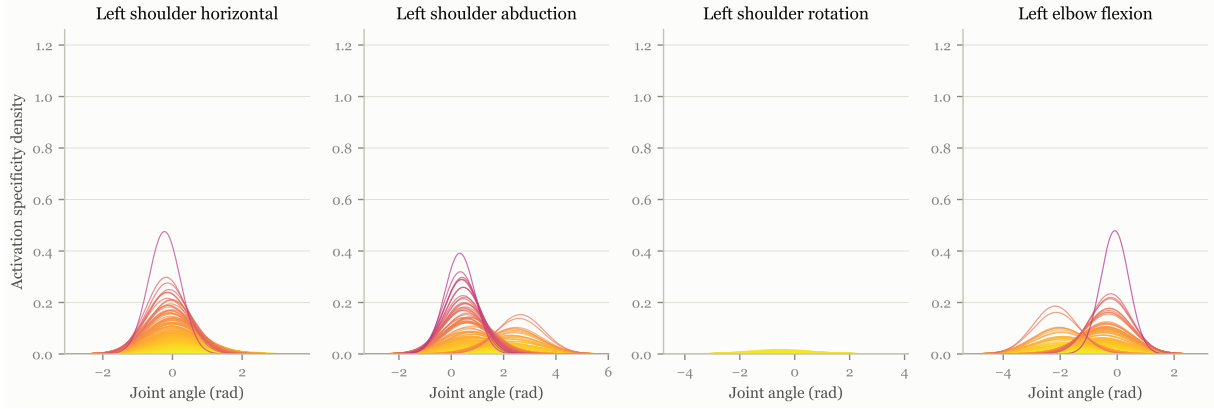


Figure 26: Activation specificity densities over the four left joint angles for **CL** at $\tau = 2.0$ and $g = 2$, drawn as in Figure 25. Note the different vertical scale.

In comparison, the densities are wider and less selective overall, and the bimodal pattern is only partially visible. **CL** has no incentive to attribute any particular meaning to a positive activation. Without a non-negativity gate, zero is an arbitrary point in the distribution of a unit, so the frames on which the unit counts as active are an arbitrary subset rather than the frames it is tuned to. This explains the worse selectivity, and the comparison favours **NNCL** by construction (Section 2.4.2.3).

The two metrics describe two different forms in which the pose is encoded. In **CL**, some units track a joint angle continuously enough to reach a decent per-unit R^2 , so the information is carried by the activation value. In **NNCL**, the non-negativity gate splits the samples into an active and an inactive set, and information is carried by whether a unit fires at all. Both forms make individual units informative about the pose, and neither is inherently better. Which form is more useful depends on the task it is put to and on how interpretability is measured.

The link between sparsity and unit-level structure extends to the temporal setting. The non-negativity gate produces units that fire selectively over narrow joint-angle ranges, as the tuning curves in Figure 25 show, demonstrating unit-level structure against continuous ground factors in the same form as found in Section 3.4 against discrete ones. **CL** encodes pose through a different mechanism: activation values track joint angles continuously, which per-unit R^2 captures but activation specificity does not. Because the two strategies make different assumptions about how information is carried, no single metric can rank them: the choice of measure does not reveal which representation is better, only which encoding strategy it is suited to.

6 Conclusion

This thesis asked whether sparsity imposed during contrastive training is sufficient to produce unit-level semantic structure, and what controls how much emerges. The question was pursued through three objectives: characterizing the range over which the non-negativity gate produces structure, separating the contribution of non-negativity from axis alignment, and testing whether the link holds in a naturalistic temporal setting.

6.1 What Was Found

The non-negativity gate reliably produces unit-level semantic structure. As the contrastive temperature increases, gate-induced sparsity rises and unit selectivity for object categories follows, measured both as neuron-level matching (Section 3.4) and as disentanglement under SEPIN (Section 3.5). This structure comes without a cost to downstream performance: identity and category probe accuracy under NNCL match those of standard CL across the stable temperature range (Section 3.6). Disentanglement also propagates to the backbone representation, suggesting that gradients flowing through the gate encourage earlier layers to adopt a more axis-aligned structure.

The non-negativity gate is conditionally stable. Three factors are found to trigger dimensional collapse: high temperatures increase the pressure to widen the similarity margin between positives and negatives beyond what the positive orthant can sustain, with a theoretical derivation of the stable range given in Section 3.2. High sparsity regularization produces the same collapse (Section 4.1), likely through a similar mechanism. Easy positive pairs are also found to contribute to collapse (Section 5.1), though the exact mechanism is less clear.

Strict non-negativity is not required to produce unit-level semantic structure. An explicit sparsity regularizer applied to standard CL recovers category matching on a par with the NNCL baseline and produces comparable disentanglement at the projector (Section 4), without deactivating a single unit. The mechanism is axis alignment: concentrating activation mass onto fewer dimensions breaks rotational invariance in the same way non-negativity does, and the same interpretability improvements follow. The non-negativity gate retains distinct advantages (marginally stronger identity matching, a higher effective rank, and units with a binary active/inactive interpretation), but these come with the conditional stability cost already noted. The two mechanisms are therefore not interchangeable. Each produces a different kind of sparsity, and each kind carries different benefits and limitations.

The link between sparsity and unit-level structure extends to the temporal setting on MIMo-Mirror (Section 5.2). Continuous ground factors complicate the comparison: both representations are shown to carry pose information, but in different forms. NNCL encodes pose through selective activation, with individual units firing over narrow joint-angle ranges. CL encodes pose through activation magnitude, with individual units showing high per-unit correlation with joint angles (Section 2.4.2.2); this likely relies on the ground factors being low-dimensional relative to the embedding width, which makes per-unit linear encoding of individual factors feasible. Neither form is strictly better: at the unit level, no shared measure captures both strategies.

6.2 What This Means

The third experiment complicates the picture. Both representations carry interpretable unit-level pose information, but the form differs between sparse and dense representations. On RT4K, where object categories are discrete, the binary nature of class assignment fits naturally into a binary encoding, and the advantage of a sparse code is correspondingly clear. On MIMo-Mirror, where ground factors are continuous, a dense representation can encode them equally well at the unit level through linear correlation. This narrows the claim: sparsity produces a binary encoding of semantic structure, and the interpretability advantage is clearest when evaluating for binary semantic structure.

Across both datasets and both sparsity mechanisms, sparsity imposed during training shapes what the encoder learns to encode. It is sufficient to produce unit-level semantic structure directly. This is reinforced by the rotational invariance argument: standard contrastive learning is rotationally invariant by construction, so individual units have no guaranteed semantic role. Both sparsity-inducing mechanisms studied here break that invariance, and both improve binary interpretability as a result.

Sparse coding in early visual cortex ([Section 1.1.5](#)) follows the same binary logic: at any moment, a small fraction of neurons are active and the remainder silent, so each neuron either participates in a representation or does not. If sparsity produces interpretable binary encodings because the constraint itself forces a binary on/off structure, then the same pressure operates in biological vision, and the interpretability advantage observed here may reflect a general property of sparse codes rather than a feature specific to contrastive learning.

6.3 Limitations and Future Work

Several questions are left open. The stability analysis identifies easy positive pairs as a contributing factor to dimensional collapse under the gate, but the exact interaction between pair difficulty, temperature, and the gradient dynamics at the positive orthant boundary is not fully characterized. Two effects are consistently observed but mechanistically unexplained: the disentanglement advantage of NNCL propagates to the backbone representation at a layer that does not interact directly with the gate, and the modest downstream accuracy advantage of NNCL first reported by Wang et al. [6] and replicated here also lacks a mechanistic account.

Several patterns within the experiments are noted but not pursued. The dip-and-recovery in CL identity matching under the explicit sparsity penalty ([Figure 21](#)) has a plausible explanation (the penalty disrupts a distributed code before axis alignment is sufficient to replace it) but is not further investigated. The decline in CL effective rank at high temperature on MIMo is consistent with the representation aligning with the subspace of the low-dimensional generating factors, but this remains a hypothesis. The adaptive relationship between object visual complexity and active-unit count ([Figure 12](#)) is visible in the data, but rests on a qualitative reading: no quantitative index of visual complexity is computed, and the variation may reflect other factors such as colour contrast or category membership.

The comparison on MIMo-Mirror rests on a single temporal dataset with a specific movement distribution, and the generalization claim is correspondingly narrow.

These open questions do not undermine the central result. Sparsity imposed during contrastive training is sufficient to produce unit-level semantic structure. The mechanism is the breaking of rotational invariance: any constraint that concentrates activation mass onto fewer dimensions removes the freedom for units to be arbitrarily rotated, and interpretability follows as a consequence.

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Statement of Resources

The following resources and tools were used in the preparation of this thesis. No other aids were used beyond those listed here.

Writing and Language Aids

TOOL	PURPOSE AND SCOPE
DeepL	Translation of individual words and phrases.
LanguageTool	Grammar and spell checking.
claude-sonnet-4-6	Revision of language and improvement of clarity in individual passages.

Software and Tools

SOFTWARE / TOOL	VERSION	PURPOSE
Python	3.12	Primary programming language.
PyTorch	2.13.0+rocm7.2	Deep learning framework.
torchvision	0.28.0+rocm7.2	Image datasets and transformations.
SciPy	1.18.0	Scientific computing and statistical analysis.
Matplotlib	3.11.0	Data visualization and plotting.
Typst	0.15.0	Thesis typesetting.
Wolfram Alpha	–	Verification of mathematical expressions, including solving derivatives.
Google Scholar	–	Literature research.

Compute Resources

RESOURCE	DESCRIPTION
Goethe-NHR cluster	High-performance computing cluster at Goethe University Frankfurt. Used for all model training experiments.

Datasets

The datasets used in this thesis are described and cited in [Section 2](#).

Bitte dieses Formular zusammen mit der Abschlussarbeit abgeben!

Erklärung zur Abschlussarbeit

gemäß § 35, Abs. 16 der Ordnung für den Bachelorstudiengang Informatik vom 17. Juni 2019:

Hiermit erkläre ich

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